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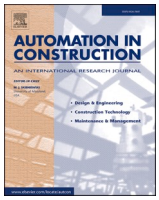

Structural defect segmentation using a semi-supervised algorithm integrating YOLO and the segment anything model

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A R T I C L E I N F O

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A B S T R A C T

Deep learning–based defect segmentation enables automated detection, classification, localisation, and quantification of structural defects. Although supervised segmentation methods yield strong performance, they require extensive pixel-level annotations and struggle with data scarcity, class imbalance, thin crack segmentation, and generalisation across diverse conditions. This paper proposes a semi-supervised two-stage segmentation framework that integrates a high-accuracy object detection model (Stage I: localisation using YOLO11/Oriented YOLO11) with zero-shot unsupervised segmentation (Stage II: pixel-level mapping using Segment Anything Model with box prompts). The method requires only object-detection labels, eliminating the need for dedicated segmentation annotations. Experiments on benchmark datasets involving steel, masonry, concrete cracks, spalling, and corrosion demonstrate that the hybrid Oriented YOLO11 and SAM model achieves state-of-the-art performance, with an average Dice score of 0.7, comparable to those of fully supervised models. The proposed framework offers real-time performance, strong generalisation, and high scalability, making it a promising solution for robust structural defect segmentation.

1. Introduction

Industrial structures, during their lifetime, are subject to adverse conditions caused by natural hazards, man-made threats, as well as several age-dependent structural performance degrading mechanisms (e.g., cracks, corrosion). Conventional visual inspection-based Structural Health Monitoring (SHM) systems often prove to be subjective and time-consuming, with difficulties in accessing inaccessible regions, hazardous areas, and high altitudes, as well as a lack of detailed results. Furthermore, the risks to life and safety inherent in traditional Structural Health Monitoring (SHM) methods significantly reduce their overall efficacy and practical benefits.

Unmanned Aerial Vehicles (UAVs) offer a valuable solution for achieving demanding Structural Health Monitoring (SHM) objectives by overcoming the limitations typically associated with conventional measurement, visual inspection, and monitoring methods. Equipped with advanced technology and sensors such as high-resolution cameras and image stabilization systems, UAVs can effectively collect and integrate data into preliminary design processes, survey construction sites, monitor and document ongoing work, and inspect and assess the condition and health of structures in hard-to-reach areas [1].

Researchers and practitioners in various fields have shown significant interest in the rapidly evolving Artificial Intelligence (AI) and computer vision (CV) technologies. Deep Learning (DL) techniques are the most popular among AI and CV technologies for achieving automated and intelligent infrastructure inspection, especially for data processing using optical data (images and videos). Deep learning has significantly enhanced the ability to robustly and accurately detect surface defects in images. Typically, deep learning applications in this domain fall into three primary computer vision tasks: image classification, object detection, and semantic segmentation. Image classification aims to categorise the type of defect present in an image, while object detection focuses on identifying both the location and category of defects using bounding boxes [2]. Segmentation aims to perform pixel-level classification of objects in the input images, allowing for the identification of defects and providing detailed information such as their location and the area covered (size).

Fully Connected Network (FCN) [3], introduced in 2014, was the first fully convolutional network trained end-to-end with supervised pre-training and fine-tuned for pixel-to-pixel predictions of defects in civil infrastructure. UNet, introduced in 2015 [4] and built upon the FCN, is a convolutional neural network designed for semantic image

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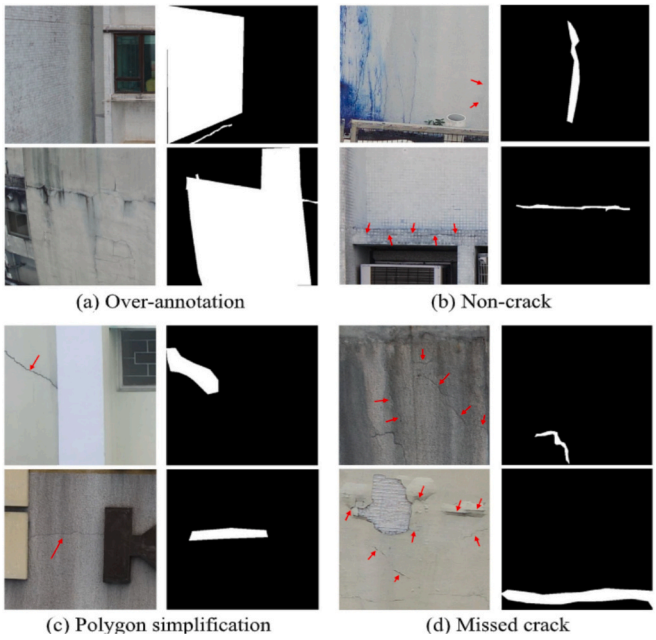
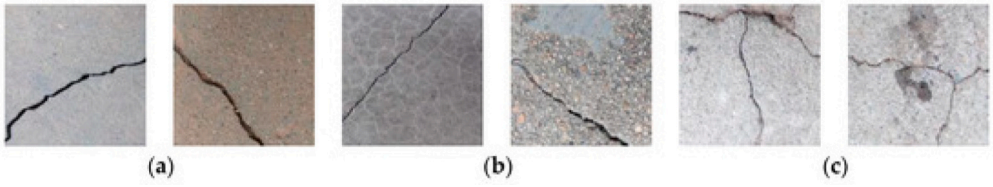
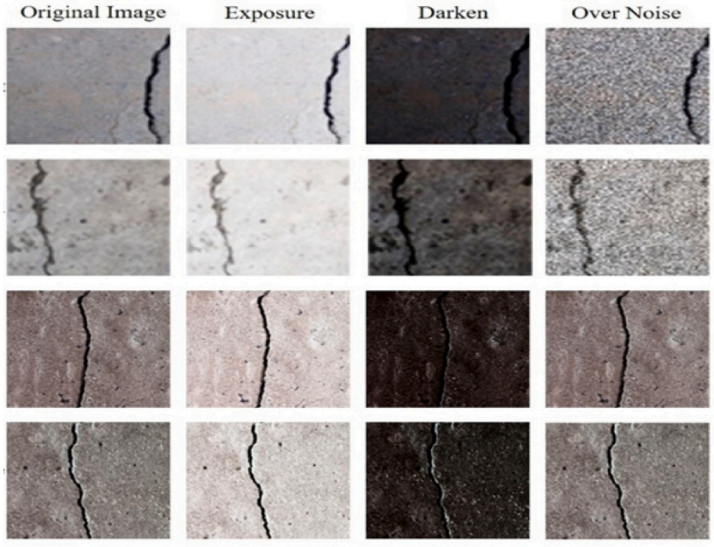


Fig. 1. Challenges in pixel-level annotation of cracks. [38].
(a) Over-annotation, (b) Non-crack, (c) Polygon simplification, (d) Missed crack.

segmentation, featuring a U-shaped architecture with an encoder (downsampling path) that extracts features and a decoder (upsampling path) that reconstructs the image. U-Net, established with a skip connection, gives pixel-level mapping of defects. Few researchers conclude that using classical image classification structures as encoders, pre-trained with data such as ImageNet [5,6], strengthens feature extraction in crack segmentation networks, enhancing crack detection performance. Han et al. [7] proposed CrackDiffusion, a two-stage framework that combines unsupervised diffusion-based crack feature extraction with supervised U-Net segmentation. However, the cold diffusion model shows limited robustness, struggling to reconstruct crack-free images when test backgrounds deviate from the training distribution. They also introduced MambaCrackNet [8], a pixel-level segmentation model integrating Visual Mamba and CNN modules to enhance fine crack detection and suppress false background responses, achieving state-of-the-art performance. Its limitation lies in the large model size and slower inference speed due to dual parallel integration, making it less suitable for mobile deployment. Zou et al. [9] proposed the RetCrack network, which utilizes a combination of multi-scale feature extraction techniques and an enhanced attention mechanism for crack detection, thereby overcoming challenges such as the loss of crack width information, limited computational efficiency, and issues with misidentification between cracked and non-cracked regions. Despite the significant success of CNN-based methods, most of these approaches have struggled to effectively model long-range dependency information [10,11,12]. Transformer models have been recently developed to address the problem of CNN-based models in segmenting long-range and complex objects in target images. Kyem et al. [11] proposed the Crack-Segmenter, which integrates three complementary modules:



(**a**) Good images, (**b**) Images with complex shading (noise), and (**c**) Images illuminated by strong light. **[20]**



(d) Images with complex backgrounds and different textures. **[21]**

Fig. 2. Challenges in images with complex shading and backgrounds.
(a) Good images, (b) Images with complex shading (noise), and (c) Images illuminated by strong light. [20]. (d) Images with complex backgrounds and different textures. [21].

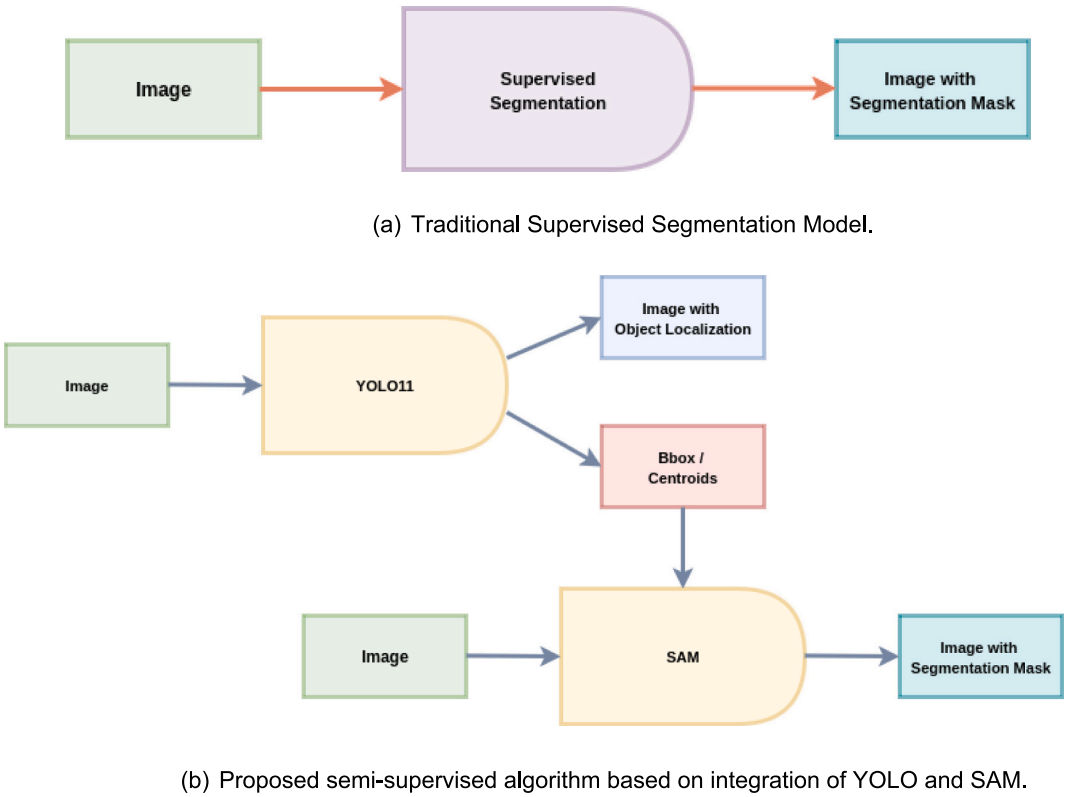


Fig. 3. Segmentation Model.
(a) Traditional supervised segmentation model. (b) Proposed semi-supervised algorithm based on integration of YOLO and SAM.

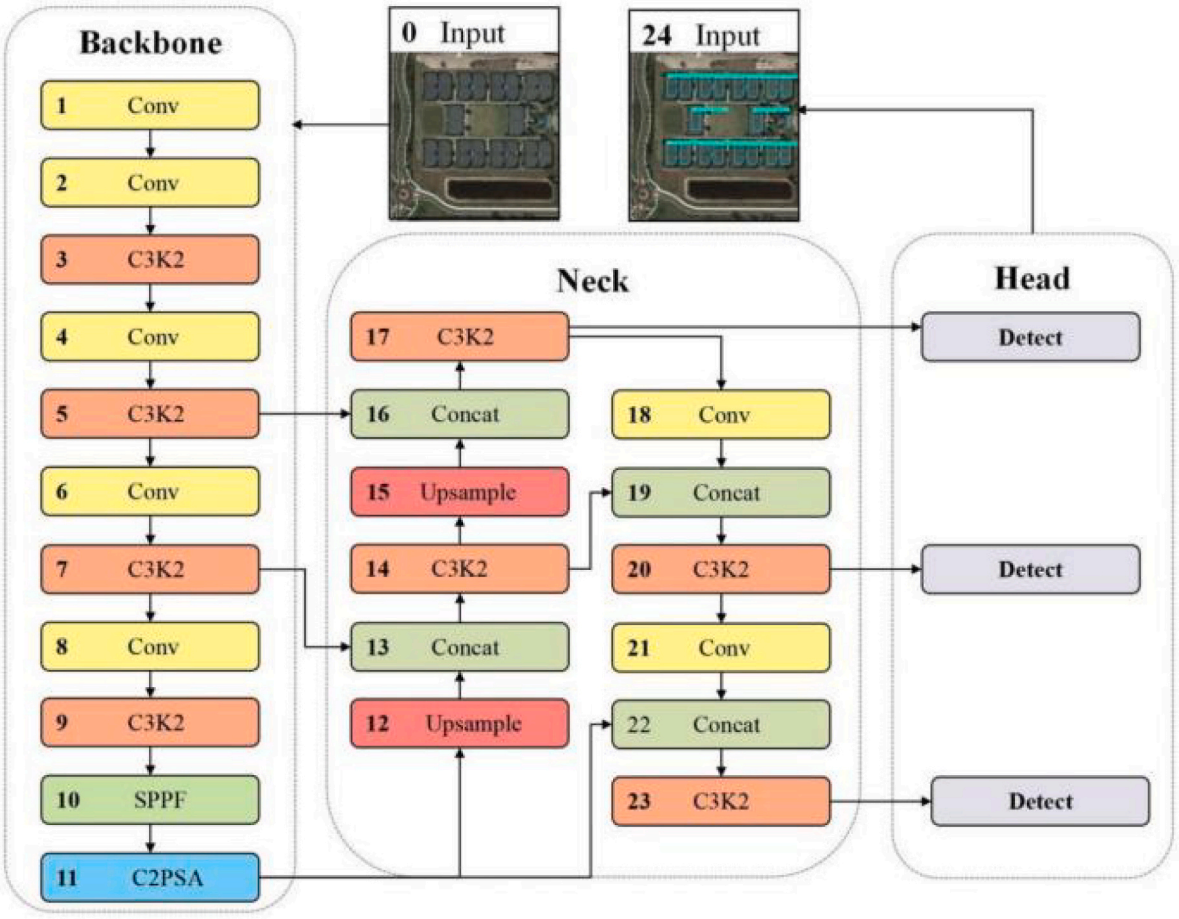
the Scale-Adaptive Embedder (SAE) for robust multi-scale feature extraction, the Directional Attention Transformer (DAT) for maintaining linear crack continuity, and the Attention-Guided Fusion (AGF) module for adaptive feature integration. Liu et al. (2021) [13] introduced CrackFormer, a transformer-based architecture specifically designed for detecting pavement cracks. The model integrates self-attention and scale-aware attention mechanisms to effectively capture both fine-scale crack patterns and long-range dependencies within the image. Building on this foundation, transformer-based frameworks have also been extended to crack detection in various other infrastructure domains, including bridges [13–14], tunnel linings [15], and wind turbine blades [16].

While both convolutional neural networks (CNNs) and Transformer-based architectures have demonstrated significant advancements in segmenting structural defects, each presents certain limitations. CNNs are inherently constrained by a limited receptive field, which restricts their capacity to model long-range dependencies and global contextual information within images [10]. On the other hand, Transformer-based models, though effective at capturing global features, often struggle to represent high-frequency image components such as fine textures and edge details. This limitation can negatively impact the detection of intricate defect patterns, such as cracks with complex geometries. Additionally, Transformer models tend to underperform when trained on small-scale datasets, which is a common challenge in structural inspection domains [17]. To overcome these challenges, there is a growing need for hybrid deep learning models that combine the local feature extraction strengths of CNNs with the global attention capabilities of Transformers. Such integration holds promise for improving segmentation performance in defect detection across diverse civil infrastructure applications.

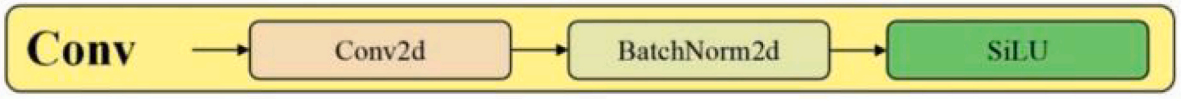
A significant limitation in the field of defect segmentation, particularly for cracks, is the lack of standardised, large-scale datasets that comprehensively capture variability across surface materials, environmental conditions, and crack morphologies. Existing datasets are limited in scope, hindering models from learning robust representations for varied real-world conditions [18–21]. To improve the generalisation ability of crack detection, research has focused on developing methods for detecting cracks in complex images. To address these challenges, researchers have developed robust methods for segmenting structural defects, which enhance the ability to distinguish between noise and actual defects. Hoskere et al. (2020) [22] introduced MaDnet (Material-and-Damage Network), a deep learning framework designed for multi-task semantic segmentation capable of identifying various types of structural damage, including cracks, spalling, and corrosion. In a related study, Bai et al. [23] assessed the effectiveness of the Mask R-CNN architecture for detecting concrete cracks and spalling. Although models like Mask R-CNN and MaDnet are effective at localisation, they are computationally demanding and not suitable for real-time applications. Researchers have employed YOLO (You Only Look Once) in real-time object detection and segmentation. While their findings demonstrated the promise of deep learning models for automated damage recognition, the evaluation was limited by the scope of the dataset. To reliably assess the model's generalisation capability, further testing on more diverse and extensive datasets is recommended.

Xu et al. [24] explored the influence of both data quality and quantity on the accuracy of crack segmentation models. Their findings highlight that training deep neural networks on limited datasets often leads to overfitting and poor generalisation performance. Improved model performance is closely tied to the availability of large, high-quality annotated datasets. However, raw imagery collected during field inspections is typically unlabeled, requiring substantial effort in data acquisition, cleaning, labeling, and curation. This process is often time-consuming, subjective, and lacks standardised annotation protocols, resulting in inconsistencies among annotators.

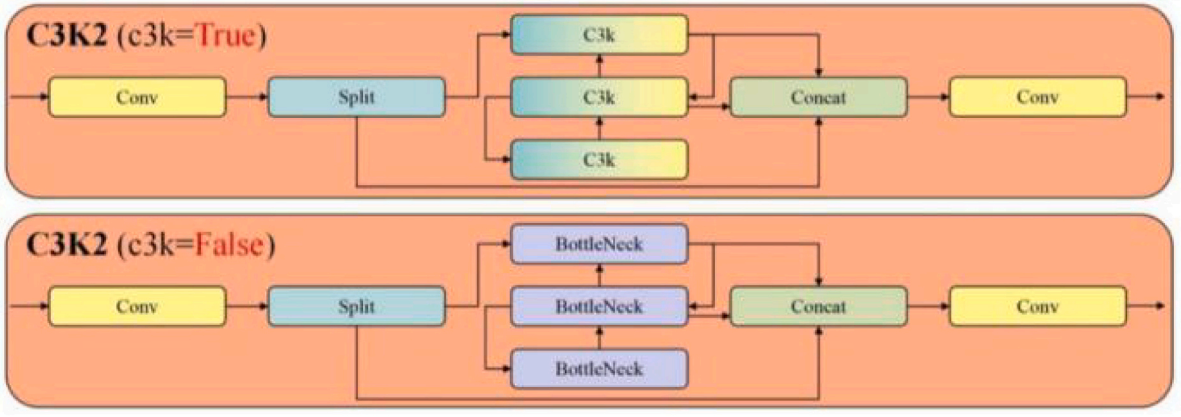
Although several open-source datasets for structural damage are available, practitioners frequently rely on data augmentation, synthetic



(a) YOLOv11 network.

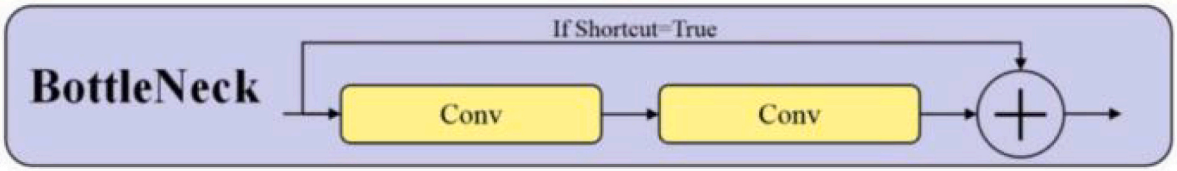


(b) Conv module.

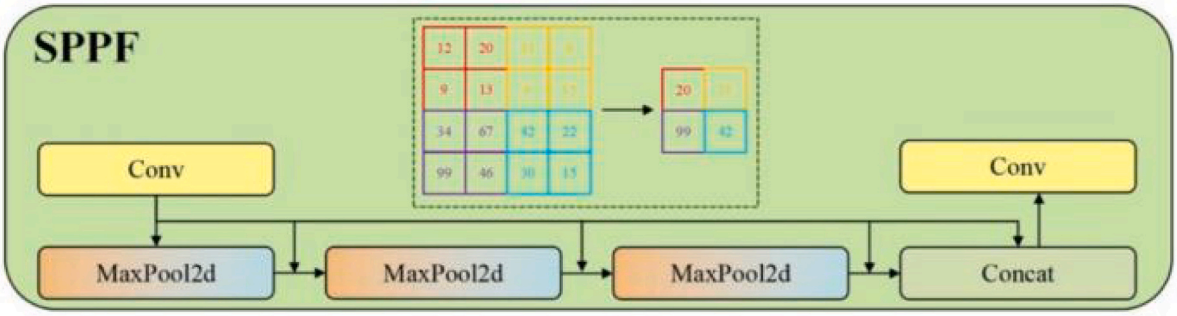


(c) C3k2 module.

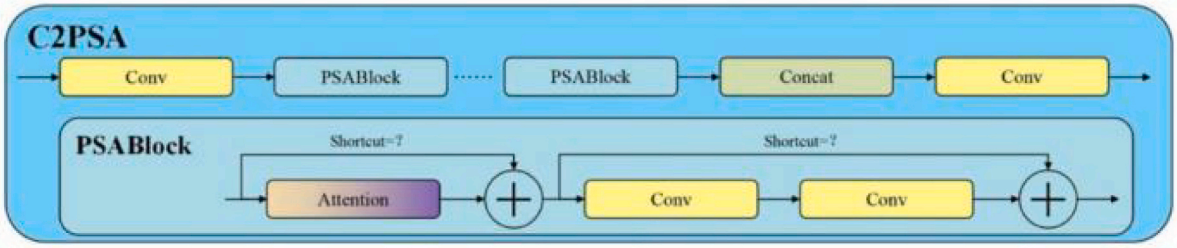
Fig. 4. YOLOv11 architecture and its modules.
(a) YOLOv11 network, (b) Conv module, (c) C3k2 module, (d) BottleNeck module, (e) SPPF module, (f) C2PSA module, (g) Detection module.



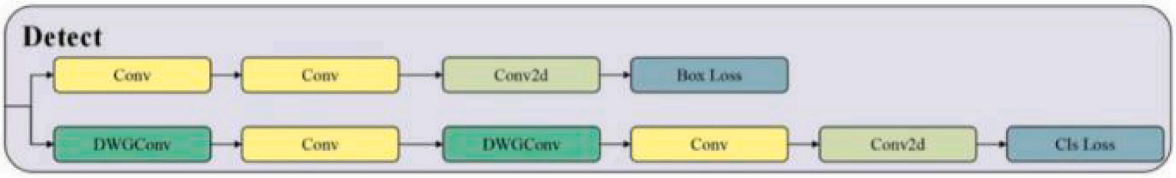
(d) BottleNeck module.



(e) SPPF module.



(f) C2PSA module.



(g) Detection module.

Fig. 4. (*continued*).

data generation, and transfer learning to expand dataset diversity and address data scarcity. Traditional data augmentation methods—such as random cropping, flipping, and noise addition—offer only marginal enhancements by introducing limited variations to the original images. Conversely, transfer learning (TL) has proven effective in low-data regimes. In this approach, pre-trained weights from a source model are used to initialize a target model, which is then fine-tuned for the specific task, improving predictive accuracy even with fewer training samples [18].

Generative Adversarial Networks (GANs), introduced by Goodfellow et al. [25], have emerged as a powerful tool for generating synthetic data in such scenarios. Unlike traditional augmentation, GANs can generate entirely new and diverse samples from random noise vectors, thus enriching the dataset with novel feature representations [26–27]. Several domain-specific GAN implementations have been proposed. For instance, Ma et al. [28] developed a GAN tailored for pavement crack augmentation to enhance the training of a YOLO-MF model. Similarly, Shin et al. [29] utilised GANs to mitigate data imbalance and scarcity in deteriorating structures, while Dunphy et al. [30] demonstrated the use of GANs for generating multi-class damage imagery and validated their utility in improving the performance of damage detection models. To address the data scarcity issue, researchers have employed data augmentation, transfer learning, and generative models to extend the dataset and achieve the required segmentation accuracy [31–32]. Recently, efforts to attain zero-shot capability in models have concentrated on a foundation model known as the Segment Anything Model (SAM) [33–35]. While SAM methods have shown effectiveness in defect detection, the learning process remains costly.

Deep learning models for segmentation tasks are critically dependent on large volumes of annotated data. In particular, segmentation networks require dense pixel-level annotations to effectively learn model parameters and adapt to diverse application scenarios [36]. However, creating such datasets remains a major challenge. High-quality annotations require expert-level domain knowledge, whereas the manual process is labour-intensive, time-consuming, and prone to inconsistencies. Even skilled annotators may struggle to accurately label defects such as cracks when faced with irregular geometries, variable widths, or complex branching patterns. Visual distractions, including stains, shadows, and surface blemishes, can cause false positives, while handwritten inspection marks often resemble cracks. Annotation inaccuracies also arise from human error, oversimplification of boundaries, or cognitive biases, resulting in noisy training datasets (Fig. 1). These

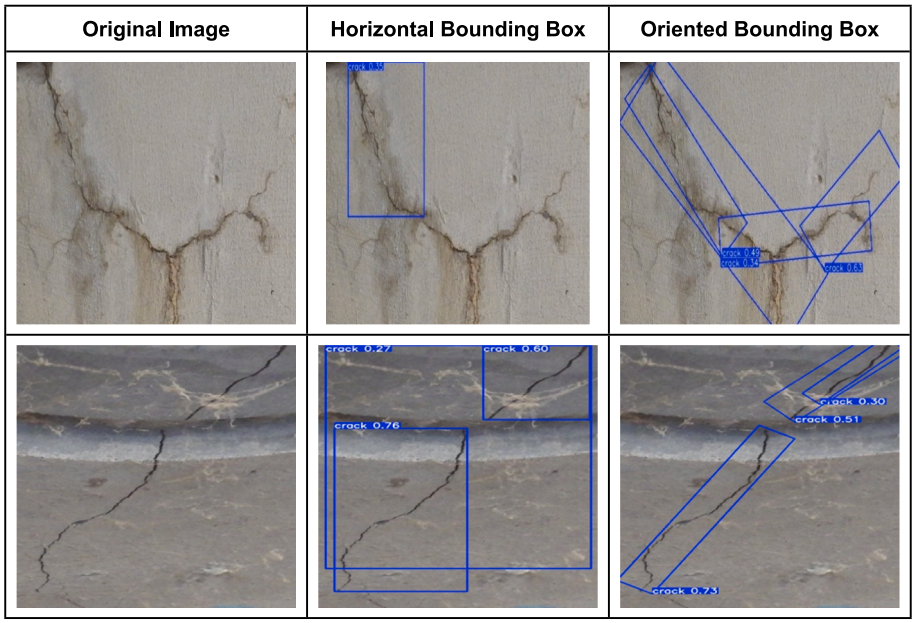


Fig. 5. Horizontal bounding boxes and oriented bounding boxes-based crack defect detection.

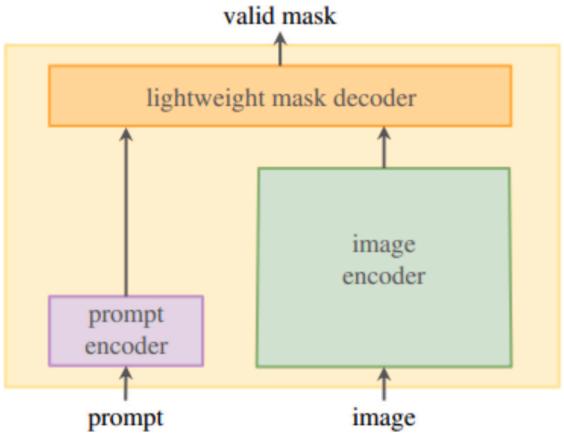


Fig. 6. SAM architecture.

challenges not only compromise model performance but can also result in false alarms and misguided maintenance actions in structural health monitoring. Furthermore, studies have shown that DL models may internalize and amplify annotation biases and errors, raising concerns about generalizability and reliability [37–38].

Defect detection becomes even more difficult under real-world conditions where structures exhibit complex backgrounds, varied lighting, low contrast, stains, and handwritten marks (Fig. 2). Conventional supervised segmentation methods, despite achieving high accuracy with extensive pixel-level training data, often fail to generalise across unseen defect types, new material surfaces, or varying environmental conditions. Their reliance on costly and time-intensive pixel-level labelling also limits scalability, rendering them impractical for large-scale monitoring of civil infrastructure.

To address these limitations, this study proposes a semi-supervised segmentation framework that integrates a high-accuracy object detection model (YOLO11/Oriented YOLO11) with zero-shot unsupervised segmentation (SAM). This two-stage pipeline enables pixel-level segmentation using only object-level annotations, eliminating the need for exhaustive pixel-wise labels. In Stage I, YOLO11/Oriented YOLO11 provides robust detection, i.e., precise bounding boxes with real-time performance. At the same time, in Stage II, SAM produces fine-grained segmentation masks in a zero-shot manner guided by prompts (bounding boxes). By combining CNN-based YOLO11 and Transformer-based SAM, the framework achieves both local feature precision and global context modelling.

A key innovation is the use of Oriented YOLO11, which generates rotated bounding boxes that align with irregular, inclined, and branching cracks. This not only improves the accuracy of crack localisation but also enables the indirect quantification of defect attributes, such as orientation, count, and length. Because the oriented bounding boxes follow the natural geometry of cracks, additional data augmentation is unnecessary, making the framework effective even with limited data. Another novelty lies in the zero-shot segmentation capability of SAM, which leverages pre-trained foundation models and semantic or region-level prompts to perform segmentation without target-specific annotations. Unlike supervised models that require retraining for each new defect type, this approach enables the introduction of new categories instantly via prompts. Different prompt types (bounding boxes, centroids, or a combination) are systematically investigated. To the best of the authors' knowledge, such prompt-based studies for structural defect segmentation have not been previously explored. The framework provides several significant advantages for civil infrastructure monitoring:

1. Reduced Annotation Effort and cost efficiency – Region-level labelling replaces exhaustive pixel-level annotation, cutting time by ~15× and cost by ~30× while avoiding over-annotation, missing defects, and noisy polygons. The reduction in annotation and retraining requirements makes this approach economically viable for large-scale monitoring programs.
2. Scalability and Generalisation – Prompt-guided SAM enhances adaptability to unseen data distributions, enabling robust performance without prior exposure. Zero-shot segmentation generalises across diverse defect types, textures, and lighting conditions, overcoming cross-dataset generalisation issues faced by supervised methods (Fig. 2).
3. Rapid Model Updates – New defect classes can be incorporated via semantic prompts without retraining, accelerating dataset updates.

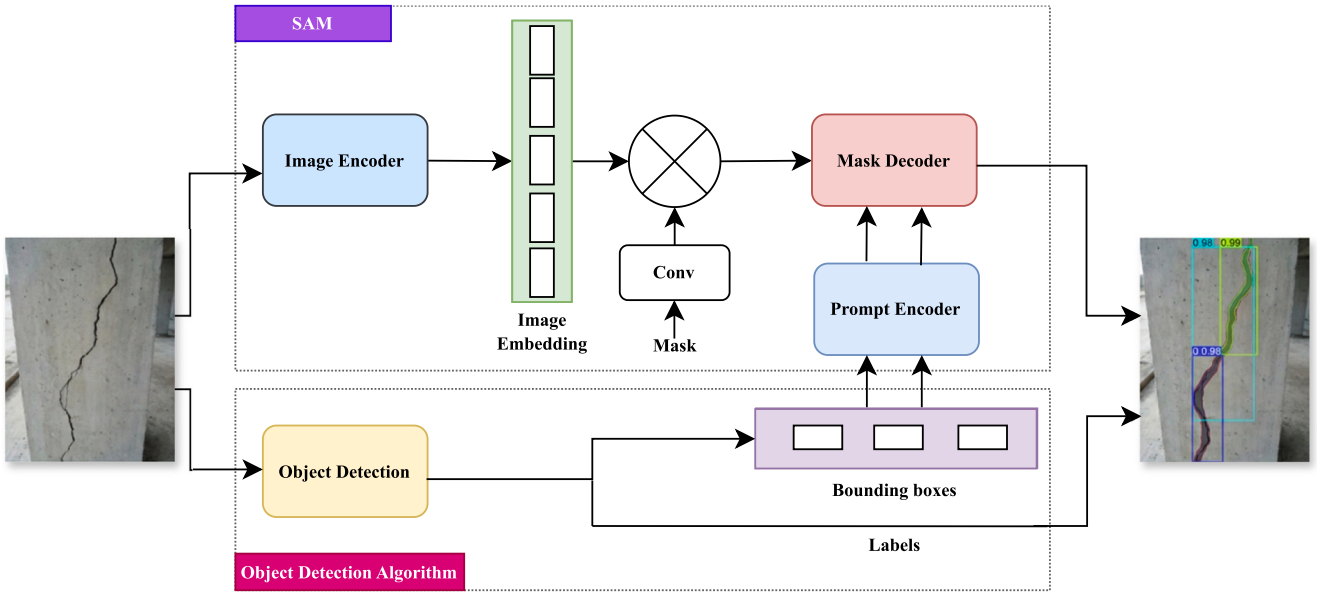


Fig. 7. Proposed YOLO11-SAM model architecture.

4. Adaptability to Real Environments – Variations in surface materials, background clutter, and environmental noise are better handled compared to pixel-supervised segmentation.
5. Human-Machine Collaboration – Region-level prompts allow inspectors to guide the segmentation process interactively, requiring far less effort than manual corrections.
6. Real-Time and Edge Deployment – YOLO11 ensures rapid detection, while SAM's flexible segmentation supports deployment on edge devices.
7. Quantitative Insights from Oriented Detection – Oriented YOLO11 bounding boxes enable approximate quantification of defect metrics (e.g., crack count, orientation, and length) directly, supporting structural condition assessment.

To the authors' knowledge, this is the first study to integrate YOLO11/Oriented YOLO11 with SAM for semi-supervised defect segmentation in civil infrastructure. The proposed framework not only reduces reliance on pixel-level annotations but also enhances generalisation, scalability, and practicality. Overall, region-level zero-shot segmentation provides a flexible, cost-effective, and adaptive solution for real-time defect detection and continuous asset management, overcoming the fundamental limitations of traditional supervised segmentation workflows.

Defect segmentation involves the precise pixel-level identification and delineation of structural flaws, such as cracks, spalling, and corrosion, enabling accurate localisation and quantification of damage. In bridge monitoring, computer vision and deep learning automates the process of detecting cracks, spalling, and corrosion for safety assessments; in buildings, it identifies façade damage, moisture ingress, and concrete degradation; in tunnels, it maps lining defects such as leakage, delamination, and deformation; and in industrial settings, it captures manufacturing defects like weld faults, coating irregularities, and material surface anomalies to improve process quality and reduce downtime. In railways, defect segmentation identifies track irregularities, sleeper degradation, and ballast fouling to enhance safety and facilitate proactive servicing. Similarly, in pipelines, it locates corrosion spots, weld discontinuities, and leaks to ensure environmental safety and uninterrupted flow [3–6]. In civil infrastructure, including bridges, highways, tunnels, and dams, these defects arise from environmental stress, load-bearing pressures, and material ageing, posing risks to safety and reliability. Automated, image-based segmentation provides objective and repeatable information, allowing engineers to monitor deterioration over time, prioritise maintenance, and implement targeted repairs. By transforming infrastructure monitoring from reactive, periodic assessments to proactive, data-driven decision-making, it enhances preventive maintenance strategies, reduces the risk of catastrophic failures, and extends the service life of critical assets while optimising inspection and repair costs. Beyond academic research, this approach supports real-world applications by automating detailed inspections, enabling early fault detection, improving operational reliability, and ensuring cost-effective asset management across diverse environments, including civil structures, pipelines, and industrial facilities.

The current study is limited to single-defect segmentation models. The remainder of this paper is structured as follows: Section 2 introduces the proposed segmentation model. Section 3 details the datasets, loss functions, and performance evaluation metrics employed in the study. Section 4 presents investigations on the proposed YOLO11 and SAM models, as well as the Oriented YOLO11 and SAM (utilising box-based prompts), under both single and cross-dataset training and testing scenarios using four benchmark crack datasets. Section 5 discusses the results obtained from the proposed YOLO11 and SAM models, along with their oriented variants, using various types of prompts. Section 6 provides a comparative analysis of the proposed segmentation framework against several widely used supervised segmentation models evaluated across the four crack datasets.

Sections 7 and 8 focus on the application of the proposed YOLO11 and SAM models for spalling with and without exposed reinforcement and metallic corrosion segmentation, including comparisons with several segmentation approaches. Section 9 concludes the paper by summarizing the key findings and outlining potential directions for future research.

2. Proposed two-stage segmentation model

The current study proposes a two-stage approach for detecting and segmenting defects, such as cracks, spalling, and metallic corrosion, in industrial structures. In the first stage, both detection and region localisation of defects is carried out using a real-time object detection model. In the second stage, unsupervised segmentation is carried out in the localized region of the defect extracted from the first stage.

The proposed two-stage approach involves combining a YOLO11/Oriented YOLO11-based object detection model and a Zero-Shot SAM

Table 1
Datasets for Evaluation.

Defect Type	Training Dataset Source and Data Augmentation details	Training Dataset Description	Data Augmentation	Testing (Testing images are completely unseen)
Crack	BPHDR (https://universe.roboflow.com/university-bswxt/crack-bphdr) Augmentation Details: Flip: Horizontal 90° Rotate: Clockwise, Counter-Clockwise, Upside Down Crop: 0 % Minimum Zoom, 30 % Maximum Zoom Rotation: Between −15° and + 15° Hue: Between −10° and + 10° Saturation: Between −25 % and + 25 % Brightness: Between −20 % and + 20 % Blur: Up to 1px	The size of images ranges from 344 × 296 to 416 × 416 Cracks are from the concrete pavement material type. Crack pixel width ranges from 1 pixel to 20 pixels Cracks are straight (horizontal or vertical), inclined, irregular, and have varied widths. Few images have good lighting conditions with adequate sunlight, and some have dark backgrounds. Other objects, such as steel beams, shadows, water stains, and tree branches occlude a few images of the cracks. The dataset is imbalanced with regard to the number of pixels representing cracks and the background	Without Augmentation: 1551 With Augmentation:4029	MCDS – 790 crack images with size of image 256 × 256 (https://doi.org/10.5281/zenodo.2601506) BCL_Steel – 2036 crack images with size of image 256 × 256 (https://doi.org/10.7910/DVN/R URXSH) BCL_Non Steel – 5769 crack images with size of image 256 × 256 (https://doi.org/10.7910/DVN/R URXSH) Training: Validation Split: 90:10
Spalling	WUTHJ(https://universe.roboflow.com/project-o6arc/spalling-wuthj) Augmentation Details: Flip: Horizontal 90° Rotate: Clockwise, Counter-Clockwise, Upside Down Crop: 0 % Minimum Zoom, 30 % Maximum Zoom Rotation: Between −15° and + 15° Hue: Between −10° and + 10° Saturation: Between −25 % and + 25 % Brightness: Between −20 % and + 20 % Blur: Up to 1px	The size of the image is 512 × 512 The dataset contains images of spalling in buildings and bridges. Images have both shallow and deep spalling Few images have spalling with and without exposed reinforcement Few images have edge spalling The spalling area ranged from 2 to 40 square pixels	Without Augmentation: 589 With Augmentation:1767	WUTHJ(https://universe.roboflow.com/project-o6arc/spalling-wuthj) Training: Validation: Testing Split: 70:15:15
Corrosion	Corrosion Condition State Semantic Segmentation Dataset (https://data.lib.vt.edu/articles/dataset/Corrosion _Condition _State_Semantic_ Segmentation_Dataset/16,624,663/2) Augmentation Details: Flip: Horizontal 90° Rotate: Clockwise, Counter-Clockwise, Upside Down Crop: 0 % Minimum Zoom, 30 % Maximum Zoom Rotation: Between −15° and + 15° Hue: Between −10° and + 10° Saturation: Between −25 % and + 25 % Brightness: Between −20 % and + 20 % Blur: Up to 1px	The size of the image is 512 × 512 The dataset considers the three categories outlined by AASHTO and BIRM standards. i. Light surface rust, exposed metal with discoloration, and freckled rust patterns were generally marked as "Fair"(Class 1). ii. Regions showing rust buildup, localized disintegration, or pack rust were labelled "Poor"(Class 2) iii. While areas exhibiting through-section damage, perforations, or structural compromise were designated "Severe"(Class 3). The distribution of instances for class 1 is approximately 950, class 2 is approximately 200, and class 3 is approximately 50.	Without Augmentation: 440 With Augmentation:1913	Corrosion Condition State Semantic Segmentation Dataset (https://data.lib.vt.edu/articles/dataset/Corrosion_Condition_State_Semantic_Segmentation_Dataset/16624663/2) Training: Validation: Testing Split: 70:15:15

All models were trained using transfer learning, initialised with pretrained segmentation weights and then fine-tuned on the target crack, spalling and corrosion datasets. The experiments are conducted on a system equipped with an Intel UHD Graphics 770 and an AMD Radeon RX 650 GPU, featuring 32 GB of RAM.

model for pixel-level defect mapping. The proposed two ensemble segmentation models are referred to as i) hybrid YOLO11 and SAM and ii) hybrid Oriented YOLO11 and SAM. The proposed two-stage approach leverages the strengths of both CNN-based models (YOLO11) and the Transformer architecture (SAM), enabling accurate and efficient segmentation with enhanced global context modelling and precise local feature preservation. The proposed ensemble approach, based on identifying and utilising bounding boxes (Stage I) as prompts for pixel-level defect mapping (Stage II), eliminates background noise and is advantageous for generalising across different datasets of similar defects. It should be noted that, instead of a bounding box predicted by the supervised object detection model from Stage I serving as the prompt for the SAM-based zero-shot segmentation in the second stage, the centroid of the bounding box, i.e., a point, and a combined point and box can also be used as prompts. The pictorial illustrations of the traditional supervised segmentation model and the proposed semi-supervised segmentation model are shown in Fig. 3(a) and Fig. 3(b), respectively.

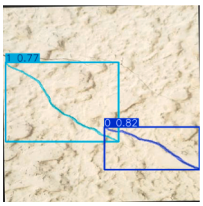
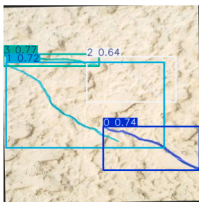
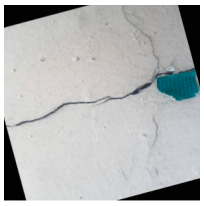
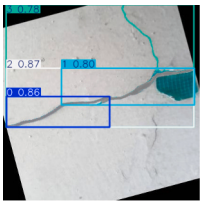
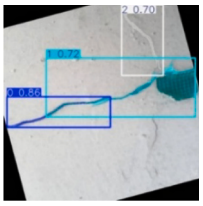
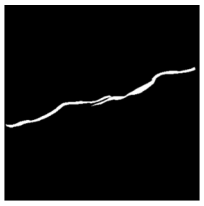
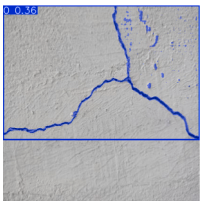
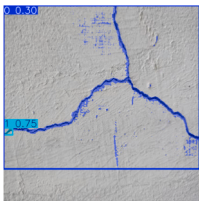
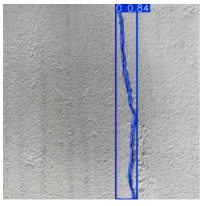
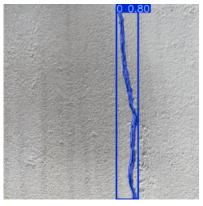
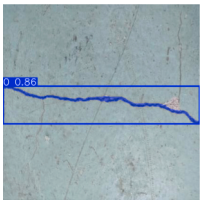
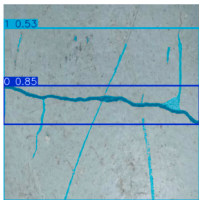
S. no	Original Image	YOLO11 and SAM	Oriented YOLO11 and SAM	GT masks	Remarks
a					Oriented YOLO11 and SAM identify four cracks. YOLO11 and SAM identify only two cracks; they miss two cracks (missing defects)
b					Both Oriented YOLO11 and SAM, as well as YOLO11 and SAM, identify all cracks, but also segment additional artefacts (misclassification).
c					Both Oriented YOLO11 and SAM, as well as YOL011 and SAM, identify all cracks; however, slight erosion is observed.
d					Both Oriented YOLO11 and SAM, as well as YOL011 and SAM, accurately identify cracks.
e					YOLO11 and SAM identify only the crack Oriented YOLO11 and SAM identify cracks but also segment artefacts, i.e. crack-like scratches.

Fig. 8. Visualisation of the prediction by the proposed semi-supervised model - Training and testing on the same dataset (BHPDR).

2.1. Stage I - classification and region mapping of defect using supervised YOLO11/oriented YOLO11 object detection model

Object detection can be viewed as a fusion of classification and localization. It involves accurately classifying and effectively locating one or more objects within an image, typically using supervised algorithms that are pre-trained on a sufficiently large, labelled training dataset [39]. The positions of the objects are localized/ determined using bounding boxes.

YOLO11 and Oriented YOLO11 are chosen as supervised object detection models in the current study. YOLOv11 uses a CSPDarknet53 backbone for efficient feature extraction, a PANet neck for multi-scale feature aggregation, and a three-headed output system (classification, regression, and objectness) for accurate object detection, all optimized for both speed and accuracy, making it ideal for real-time applications.

2.1.1. YOLO11 architecture

The YOLO family of models revolutionised object detection by unifying bounding box regression and classification within a single neural network, enabling end-to-end differentiable training and real-time inference. At its core, YOLO architectures share three main components:

Backbone: the feature extractor that converts raw images into multiscale feature maps.

Neck: an intermediate processing stage that aggregates and refines features across scales.

Head: the prediction stage that outputs bounding box locations, object classes, and confidence scores.

YOLOv11 preserves this fundamental structure but incorporates several notable innovations compared to earlier versions (e.g., YOLOv8). These include the incorporation of the C3K2 block, C2PSA attention module, and optimized detection mechanisms, which together improve speed, accuracy, and adaptability across diverse detection tasks.

Backbone:

The backbone in YOLOv11 is designed to refine feature

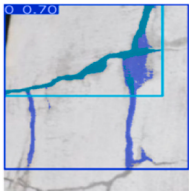
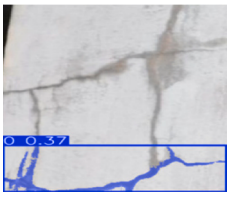
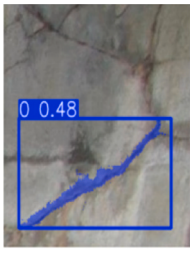
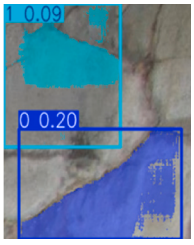
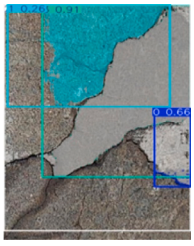
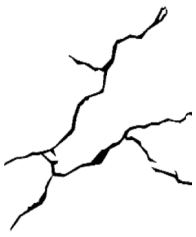
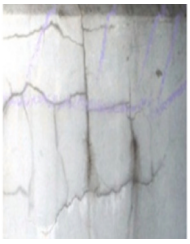
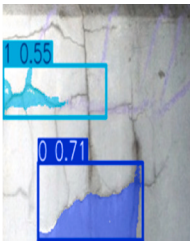
S. no	Original Image - MCDS dataset	YOLO11 and SAM	Oriented YOLO11 and SAM	GT Masks	Remarks
a					Oriented YOLO11 and SAM **miss** cracks due to complex background, lighting and noisy images, and the irregular and alligatory nature of the crack. YOLO11 and SAM identify mostly cracks, but **miss** cracks at the edges of the image
b					Both Oriented YOLO11 and SAM and YOLO11 and SAM models **miss** cracks due to complex background, lighting and noisy images, and the irregular and alligatory nature of the crack.
c					Both Oriented YOLO11 and SAM, as well as YOLO11 and SAM models **miss** cracks due to complex background, lighting and noisy images, and the irregular and alligatory nature of the crack.
d					Both Oriented YOLO11 and SAM, as well as YOLO11 and SAM models **miss** cracks due to complex background, lighting and noisy images, and the irregular and alligatory nature of the crack.

Fig. 9. Visualisation of the prediction by the proposed semi-supervised model - Training on BHPDR and testing on MCDS.

representations through multiple lightweight yet expressive modules progressively. It integrates:

Conv module: A sequence of Conv2D, BatchNorm2D, and SiLU activation layers that extract low-level spatial patterns. SiLU activation enhances gradient flow and feature expressiveness compared to the ReLU activation function.

C3K2 module: Unlike YOLOv8, which primarily employed C2f blocks, YOLOv11 introduces the C3K2 block. It employs smaller convolutional kernels and multi-branch pathways to strike a balance between computational efficiency and strong feature representation. Depending on task requirements, the block can operate in two modes:

c3k = True: adopts a multi-branch C3 structure prioritizing efficiency and lightweight processing.

c3k = False: adopts a deeper bottleneck structure for tasks requiring higher accuracy.

SPPF module: The Spatial Pyramid Pooling - Fast block retains its role from earlier versions but is optimized for real-time detection by pooling features at multiple receptive fields to capture both local and global context.

C2PSA module (new in YOLOv11): Introduces cross-scale pixel spatial attention, enabling the network to emphasise critical target regions while suppressing irrelevant background information. This module allows the model to capture finer details more effectively and handle occlusion, a limitation in previous versions.

Neck:

The neck integrates and refines multiscale features generated by the backbone. Similar to earlier YOLO designs, it employs upsampling and feature concatenation to combine semantic context with spatial detail. However, YOLOv11 differentiates itself through two enhancements.

C3K2 Integration: By replacing C2f with C3K2 in the neck, YOLOv11 improves the speed–accuracy balance during feature aggregation.

Attention Mechanism: Integrated spatial attention using C2PSA, distinguishing YOLOv11 from YOLOv8 by improving accuracy on small or partially occluded objects.

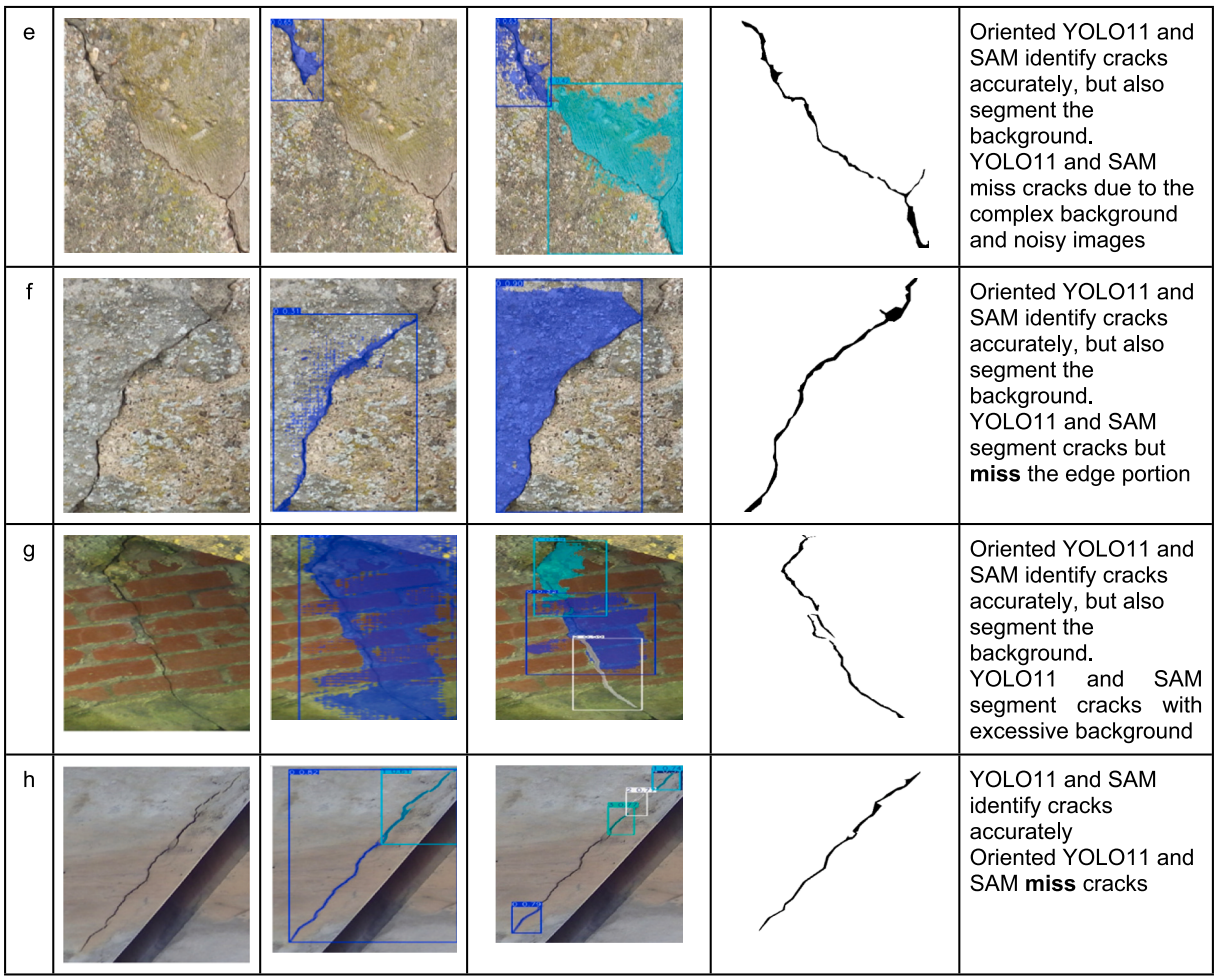


Fig. 9. (*continued*).

Head:

The head transforms aggregated features into final detection outputs, including bounding box coordinates, class probabilities, and objectness scores. YOLOv11 retains the multi-branch design of previous YOLO versions, but introduces refined processing blocks for enhanced flexibility.

C3K2 in the Head: Multiple C3K2 blocks are employed at different depths, allowing the head to capture objects across scales more effectively.

CBS Blocks: Convolution-BatchNorm-SiLU layers follow the C3K2 modules, stabilizing the data flow and enhancing nonlinear feature representation.

Final Layers and Detection: Each detection branch concludes with Conv2D layers, followed by the Detect module, which generates bounding box regression, class predictions, and confidence scores. Non-Maximum Suppression (NMS) is then applied to filter redundant detections.

Compared to YOLOv8, the combination of C3K2 with CBS blocks and C2PSA in the head allows YOLOv11 to balance computational efficiency with representational strength, achieving more accurate detection across scales and scenes.

Detection Module.

The detection module utilizes depth wise separable convolutions to significantly reduce the number of parameters and computational cost, while employing a multitask separation strategy for bounding box regression and classification. This design enhances detection accuracy and efficiency, enabling lightweight yet robust performance.

Differences from previous YOLO versions.

YOLOv11 builds upon YOLOv8 while addressing its limitations through:

1. C3K2 Block vs. C2f Block (Backbone): Greater efficiency and flexibility in feature extraction, offering faster computation without sacrificing accuracy.
2. Incorporation of C2PSA Attention (Backbone): Enhances focus on critical image regions, particularly for small and occluded objects, absent in YOLOv8.
3. Improved Neck Integration: Replacing older modules with C3K2 and integrating cross-scale attention via C2PSA for more effective feature fusion.
4. C3K2 and CBS Blocks (Head): Incorporates C3K2 and CBS blocks for multi-path feature refinement. These blocks provide flexibility with lightweight and high-representation modes (via c3k settings).
5. Lightweight yet Accurate Detection: Depthwise separable convolutions and modular task separation in the Detect head yield higher speed and precision compared to prior versions.

The YOLO11 architecture and its modules are shown in Fig. 4.

2.1.2. *Oriented YOLO11*

YOLO11 considers the bounding box to be rectangular, aligned with the image's axis. The bounding boxes of YOLO11 are referred to as Horizontal Bounding Boxes (HBB). HBBs often encompass a significant portion of the background in the box, in case of rotated or irregularly shaped objects (defects). In other words, the HBB can include large areas of irrelevant regions, which may impact the accuracy and effectiveness

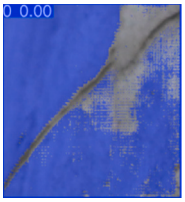
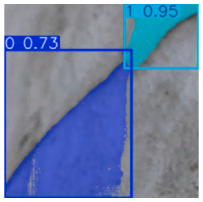
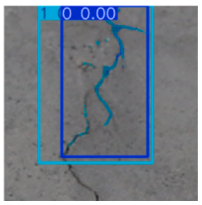
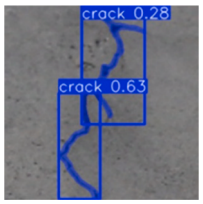
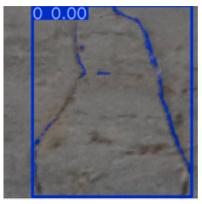
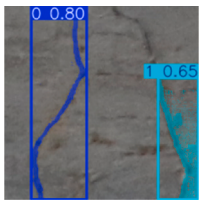
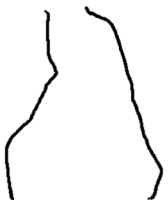
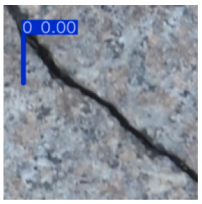
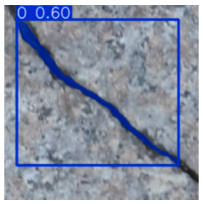
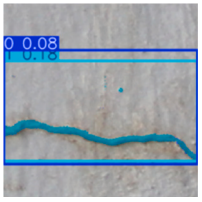
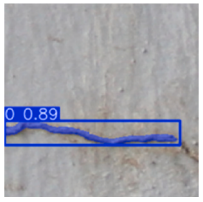
S. no.	Original Image - BCL_non steel	YOLO11 and SAM	Oriented YOLO11 and SAM	GT masks	Remarks
a					Both Oriented YOLO11 and SAM, as well as YOLO11 and SAM, accurately identify the crack and also segment the background near the crack within the bounding box.
b					Oriented YOLO11 and SAM identify multiple cracks accurately. YOLO11 and SAM **miss** one crack.
c					Both Oriented YOLO11 and SAM, and YOLO11 and SAM **miss** one crack.
d					Oriented YOLO11 and SAM identify two cracks but **miss** the edge portion due to poor bounding box prediction. YOLO11 and SAM **miss** cracks.
e					Both Oriented YOLO11 and SAM, as well as YOLO11 and SAM, identify the crack; however, Oriented YOLO11 and SAM miss the edge portion due to poor bounding box prediction.

Fig. 10. Visualisation of the prediction by the proposed semi-supervised model - Training on BHPDR and testing on BCL_non steel crack.

of the model. Horizontal Bounding Boxes (HBBs) often result in excessive overlaps and include large background regions, which hinder the effective interpretation and adjustment of detection results [40] and require post-processing steps, such as non-maximum suppression, to remove redundancy. It is unsuitable for objects with arbitrary orientations, such as rotated or skewed structures (inclined cracks) [41], as it fails to consider the orientation of the objects. However, YOLO11 requires only horizontally annotated images, which is comparatively less labour-intensive to prepare in case of custom datasets. The drawback of YOLO11 is that it doesn't consider the orientation of cracks and does not provide any information about the crack length. To improve the detection of oriented cracks by YOLO11, the dataset should include image augmentations involving rotations, which would, in turn, increase the training time and cost.

The cracks in real structures are often irregular, inclined and the direction/orientation changes as the crack propagates. When the crack growth direction changes frequently, determining the orientation of sub-cracks becomes challenging, complicating the development of regression functions to predict their orientation. Instead of horizontal bounding boxes, rotated bounding boxes is a promising alternative to overcome the above challenges.

Oriented YOLO11 introduces the capability to detect objects with an orientation angle, providing more accurate localisation of rotated objects [42] using Oriented/Rotated Bounding Boxes. Oriented Bounding Boxes (OBBs) enhance object detection precision and facilitate orientation recovery through image warping, which is especially effective for aerial images of structural defects where objects may be rotated and not aligned with the image axes. OBBs are better suited for objects with arbitrary orientations, offering a more accurate representation of object boundaries. Additional to standard rectangular bounding boxes, OBBs also capture the angle between the edge (long or short) of the BB and the x-axis [43]. Oriented YOLO11/YOLO11-OBB is particularly beneficial for detecting oriented cracks. Oriented YOLO11 employed an oriented bounding box aligned with the orientation of the crack. Comparatively,

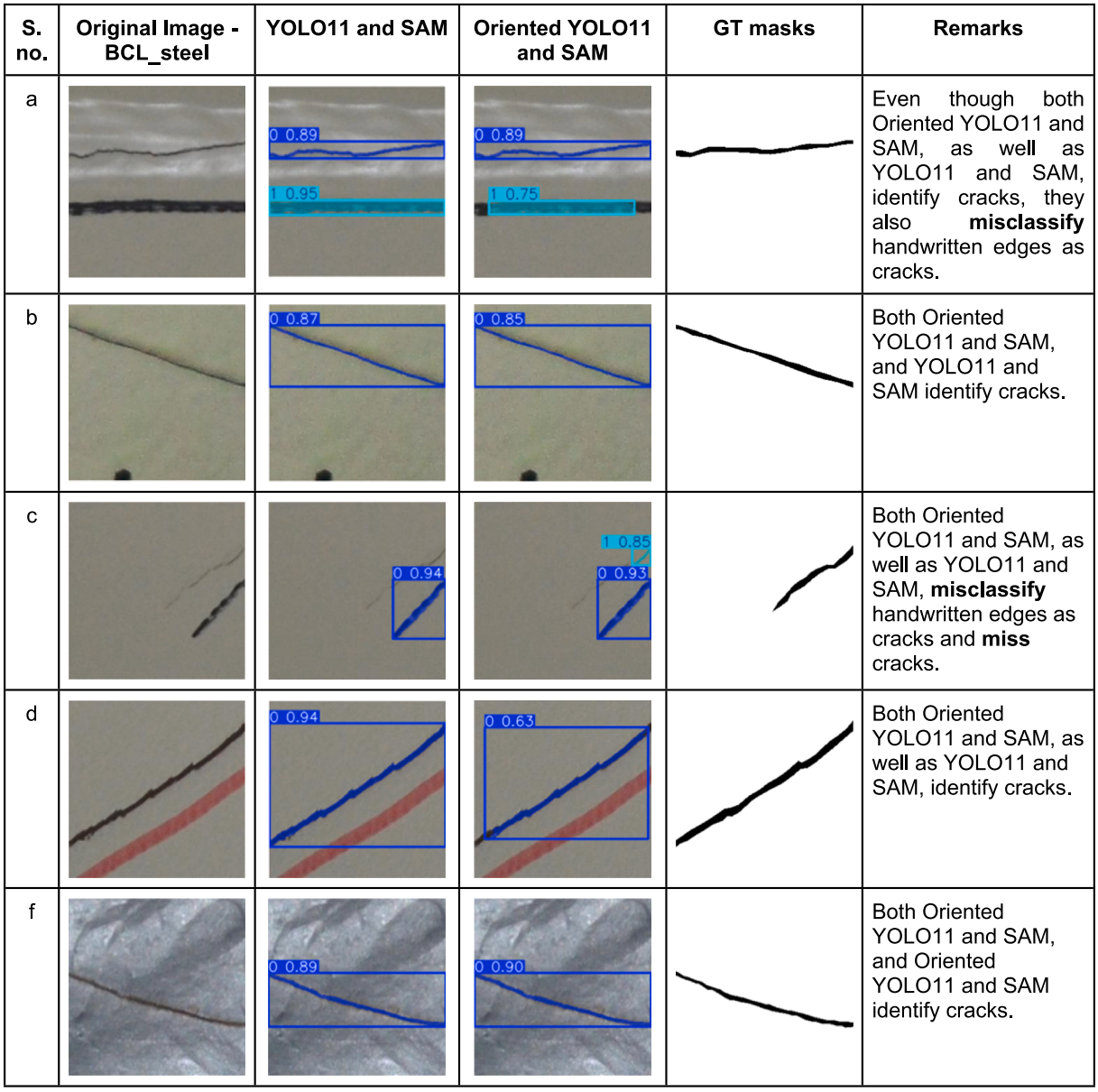


Fig. 11. Visualisation of the prediction by the proposed semi-supervised model - Training on BHPDR and testing on BCL_steel crack.

it doesn't explicitly require image augmentations involving rotations, which significantly reduced the volume of training data and time required. YOLO-OBB provides the orientation and length of the crack, enabling more precise localisation and characterisation of the cracks within the image. Due to the reduced or no requirement for data augmentation or a reduction in the volume of training data, the computational time and power associated with training any supervised model decrease.

The comparison of horizontal and oriented bounding boxes of crack defects predicted by YOLO11 and Oriented YOLO11 object detection models for irregular crack defects is shown in Fig. 5. It can be observed from Fig. 5 that the Oriented YOLO11 effectively identifies the region of inclined cracks compared to YOLO11.

2.2. Stage II - pixel-level mapping of defect using zero shot segmentation model

SAM (Segment Anything Model) is an unsupervised segmentation model. The zero-shot segmentation technique enables the model to segment objects in an image without requiring it to be specifically trained on those object categories, based on prompts such as points, boxes, or masks [44–47]. The SAM basically outputs the pixel-level mapping of the object, with the input being the image and the prompt.

2.2.1. SAM architecture

To achieve zero-shot generalisation in image segmentation, Kirillov et al. [33] introduced the concept of *promptable segmentation*. In this framework, given an image and a user-defined or automatically generated prompt, the model aims to produce a corresponding segmentation mask. A prompt essentially defines the target object to be segmented by providing spatial or semantic cues, which may originate from manual input (e.g., clicks or bounding boxes) or automated detection models.

The architecture designed to support this task, illustrated in Fig. 6, consists of three major components:

Image Encoder: The input image is first processed by an image encoder, which converts it into a high-dimensional embedding that captures its visual characteristics. The encoder operates once per image, allowing efficient reuse across multiple prompts. It is implemented using

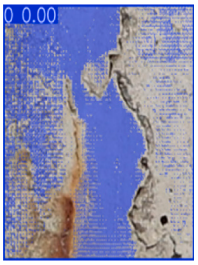
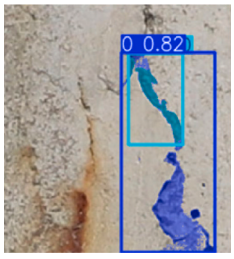
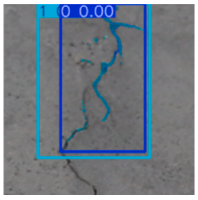
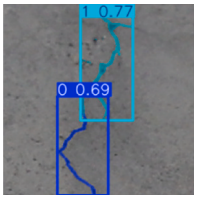
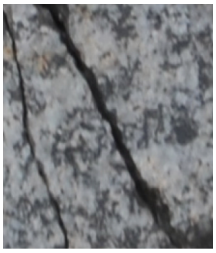
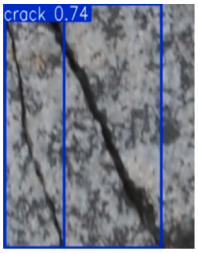
S. no	Original Image - BCL_non steel	YOLO11 and SAM	Oriented YOLO11 and SAM	GT masks	Remarks
a					Oriented YOLO11 and SAM identify two cracks, but miss one crack. YOLO11 and SAM were unable to segment/ miss the crack due to the complex, noisy background, which included shadows and stains.
b					Oriented YOLO11 and SAM identify multiple cracks accurately. YOLO11 and SAM **miss** one crack.
c					Oriented YOLO11 and SAM identify two cracks, but **miss** the edge portion of the crack due to poor bounding box prediction. YOLO11 and SAM **miss** cracks, despite the bounding box being predicted by YOLO11, due to the noisy and complex background.

Fig. 12. Visualisation of the prediction by the proposed semi-supervised model - Training on MCDS and testing on BCL_non steel crack.

a pre-trained Vision Transformer (ViT), available in different configurations—ViT-H (huge), ViT-L (large), and ViT-B (base)—that vary in size and computational complexity.

Prompt Encoder: This component translates prompts into vector representations compatible with the image embedding. Sparse prompts, such as points and bounding boxes, are encoded using positional embeddings; text prompts are processed with CLIP [48] and mask prompts are handled through convolutional layers. Point inputs are provided as pixel coordinates, labelled as either foreground or background, while bounding boxes are defined by their corner coordinates.

Mask Decoder: The decoder combines the embeddings produced by the image and prompt encoders to generate segmentation masks. This module is lightweight and can be fine-tuned for specific tasks, allowing for adaptation to various segmentation applications.

SAM offers two operating modes. In the *automatic mode* ("everything mode"), the model generates masks for all detectable objects within an image without requiring additional input. In interactive mode, the user provides prompts, such as clicks or bounding boxes, to guide the model in segmenting specific regions of interest. The current study employs the bounding box/point inside the bounding box/both predicted by the YOLO11 model as a prompt.

To the best of the author's knowledge, traditional box-based prompts are commonly used with SAM. However, there is a lack of systematic studies considering the possible different types of prompts, such as point-based, bounding box-based, mask-based, or hybrid approaches, and no established guidelines currently exist to determine the most effective prompt type for civil defect segmentation.

Researchers employ different types of prompts of SAM: (i) Point prompt (user-defined) (ii) Bounding Box prompt (iii) Auto prompt (automatic point/grid of points) [44–47].

The current study proposes to employ the following:

(i) Point-based prompt where the centroid of the bounding box is automated as the point. For the case where, multiple objects are within a bounding box, point prompts can specify which object should be segmented. Additionally, it also avoids irrelevant regions within the bounding boxes.
(ii) bounding box-based prompt.
(iii) Combination of point and bounding box prompt: The use of a combination of point and bounding box prompts is developed in the current study, keeping in mind the change in the orientation of the crack as it propagates, and an irregular, complex pattern of the cracks. Different types of prompts will be evaluated for crack defects on a large number of diverse datasets to improve accuracy in pixel-level mapping, as discussed in the subsequent section. The combination of a supervised object detection model from Stage I and its integration with SAM with different types of prompts in Stage II, results in the following six semi-supervised segmentation models.
i) Hybrid YOLO11 and SAM with box-based prompt.
ii) Hybrid YOLO11 and SAM with point-based prompt.

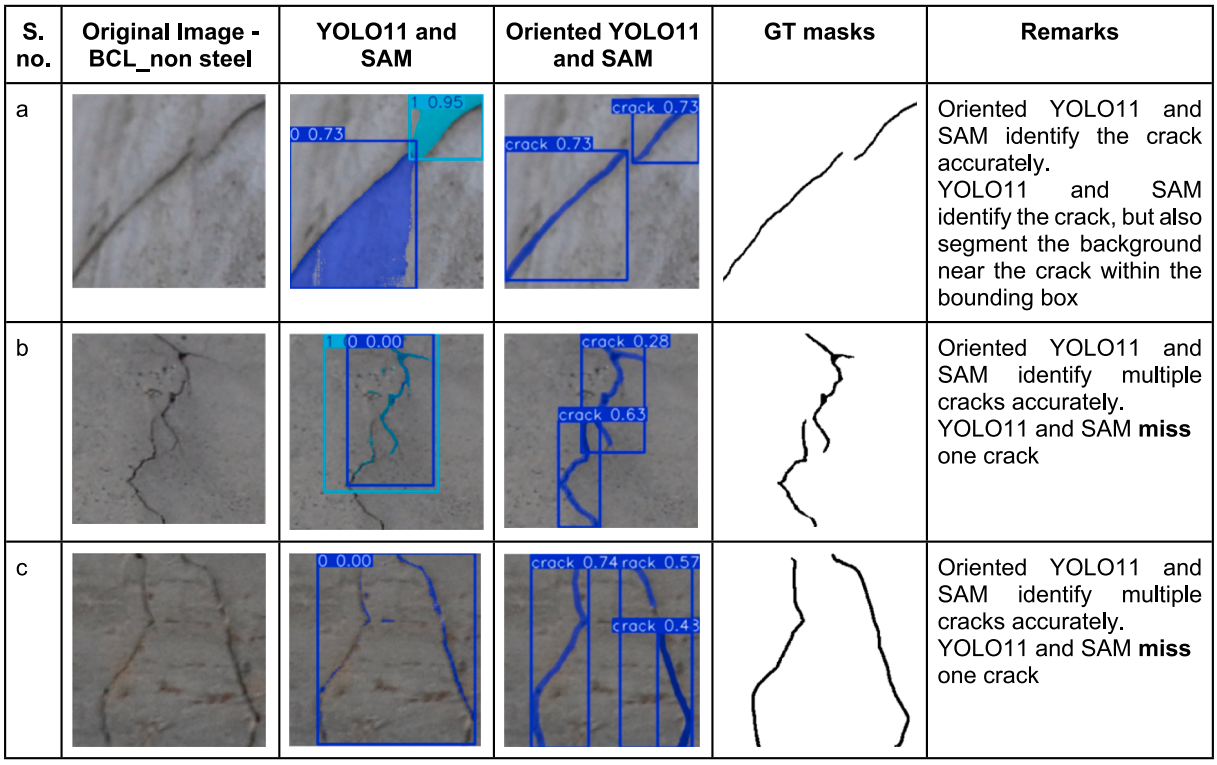


Fig. 13. Visualisation of the prediction by the proposed semi-supervised model - Training on BCL_steel, MCDS and BHPDR and testing on BCL_non-steel crack.

Table 2
Performance metrics of the proposed hybrid YOLO11 and SAM, and hybrid Oriented YOLO11 and SAM with single and cross-dataset training and testing.

Sl. No.	Scenario	YOLO11 and SAM	Oriented YOLO11 and SAM
1	Train with BPHDR and test on BPHDR	Avg Dice Score (DS): 0.6954 Highest DS: 0.9410 95 % CI: [0.638 0.789] mIOU: 0.533	Avg DS: 0.6710 Highest DS: 0.9313 95 % CI: [0.578 0.784] mIOU: 0.505
2	Train with BPHDR and test on MCDS	Avg DS: 0.3870 Highest DS: 0.7951 95 % CI: [0.241 0.543] mIOU: 0.612	Avg: 0.2004 Highest DS: 0.5692 95 % CI: [0.169 0.332] mIOU: 0.111
3	Train with BPHDR and test on BCL_non steel	Avg DS: 0.6000 Highest DS: 0.9063 95 % CI: [0.515 0.785] mIOU: 0.430	Avg DS: 0.5715 Highest DS: 0.8595 95 % CI: [0.492 0.751] mIOU: 0.403
4	Train with BPHDR and test on BCL_steel	Avg: 0.6319 Highest DS: 0.8952 95 % CI: [0.559, 0.725] mIOU: 0.462	Avg DS:0.6463 Highest: 0.8874 95 % CI: [0.580, 0.733] mIOU:0.478
5	Train with MCDS and test on BCL_non steel	Avg DS: 0.6012 Highest DS: 0.8904 95 % CI: [0.521, 0.712] mIOU: 0.435	Avg DS: 0.5468 Highest DS: 0.8562 95 % CI: [0.475, 0.682] mIOU: 0.388
6	Train with BHPDR, MCDS and BCL_steel and test on BCL_non steel	Avg: 0.7951 Highest: 0.9236 95 % CI: [0.760, 0.851] mIOU:0.662	Avg: 0.6819 Highest: 0.9552 95 % CI: [0.626, 0.795] mIOU: 0.518

iii) Hybrid YOLO11 and SAM with a combined point and box-based prompt.
iv) Hybrid Oriented YOLO11 and SAM with box-based prompt.
v) Hybrid Oriented YOLO11 and SAM with point-based prompt.
vi) Hybrid Oriented YOLO11 and SAM with combined point and box-based prompt.

2.3. *Integrated YOLO11-SAM architecture*

The integrated YOLO11-SAM architecture is shown in Fig. 7. The integrated procedure of YOLO11/Oriented YOLO11 and SAM for segmentation is as follows.

Step 1: Feature Extraction:

The images are pre-processed to remove noise and enhance texture quality. Deep convolutional layers (backbone of YOLO11) are then employed to extract both low-level surface irregularities and high-level structural defect features.

Step 2: Real-Time Detection with YOLOv11:

YOLOv11 processes the extracted feature maps to detect potential defect regions. Through anchor-based bounding box predictions, the model achieves accurate localisation of defects with varying shapes and sizes.

Step 3: Fine-Grained Segmentation with SAM.

The detected regions of interest are further analysed using the Segment Anything Model (SAM). This step performs pixel-level segmentation, producing precise masks that capture the exact boundaries of cracks, spalling, or other surface defects.

Step 4: Integrated Prediction.

Finally, the bounding box detections from YOLOv11 are combined with the detailed masks generated by SAM, resulting in a unified segmentation output that ensures both accurate defect localisation and boundary delineation.

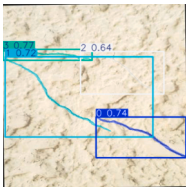
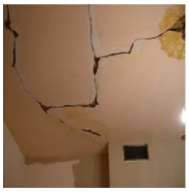
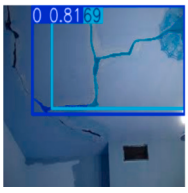
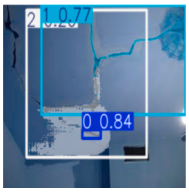
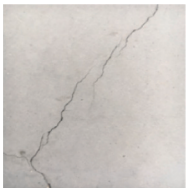
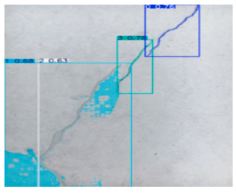
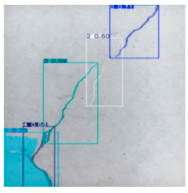
Original Image	YOLO11 and SAM	Oriented YOLO11 & SAM	GT masks	Remarks
				YOLO11 and SAM have missed detecting 2 out of 4 cracks; Oriented YOLO11 and SAM have detected all 4.
				YOLO11 and SAM have missed some parts of the cracks; Oriented YOLO11 and SAM have detected almost the whole.
				YOLO11 and SAM have missed detecting one crack out of 5; Oriented YOLO11 and SAM have detected all 5 cracks.

Fig. 14. Box-based prompt.

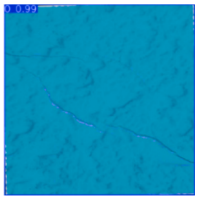
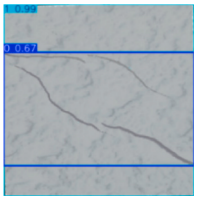
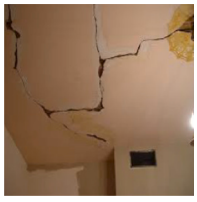
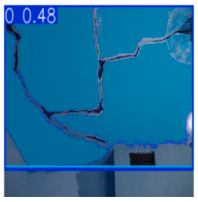
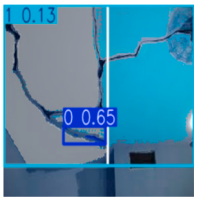
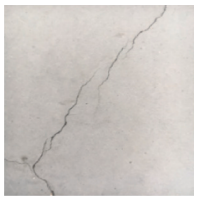
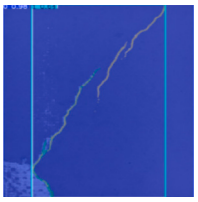
Original Image	YOLO11 and SAM	Oriented YOLO11 and SAM	GT masks	Remarks
				The cracks have not been segmented precisely; A lot of erosion is observed.
				The cracks have not been segmented precisely; A lot of erosion is observed.
				The cracks have not been segmented precisely; A lot of erosion is observed.

Fig. 15. Points-based prompt –centroids of bounding box.

3. Datasets and performance evaluation metrics

3.1. Dataset description

Three types of structural defects are considered, namely i) cracks in concrete, steel, masonry, pavement, etc., ii) concrete spalling with and without reinforcement, and iii) metallic corrosion. The dataset details are given in Table 1. Four crack datasets are used in the study. The images across the four crack datasets are varied in dimensions, as well as in surface textures, lighting, and crack morphologies. These four crack datasets are considered to evaluate the proposed segmentation model in both homogeneous ("within dataset") and heterogeneous ("across dataset") dataset segmentation.

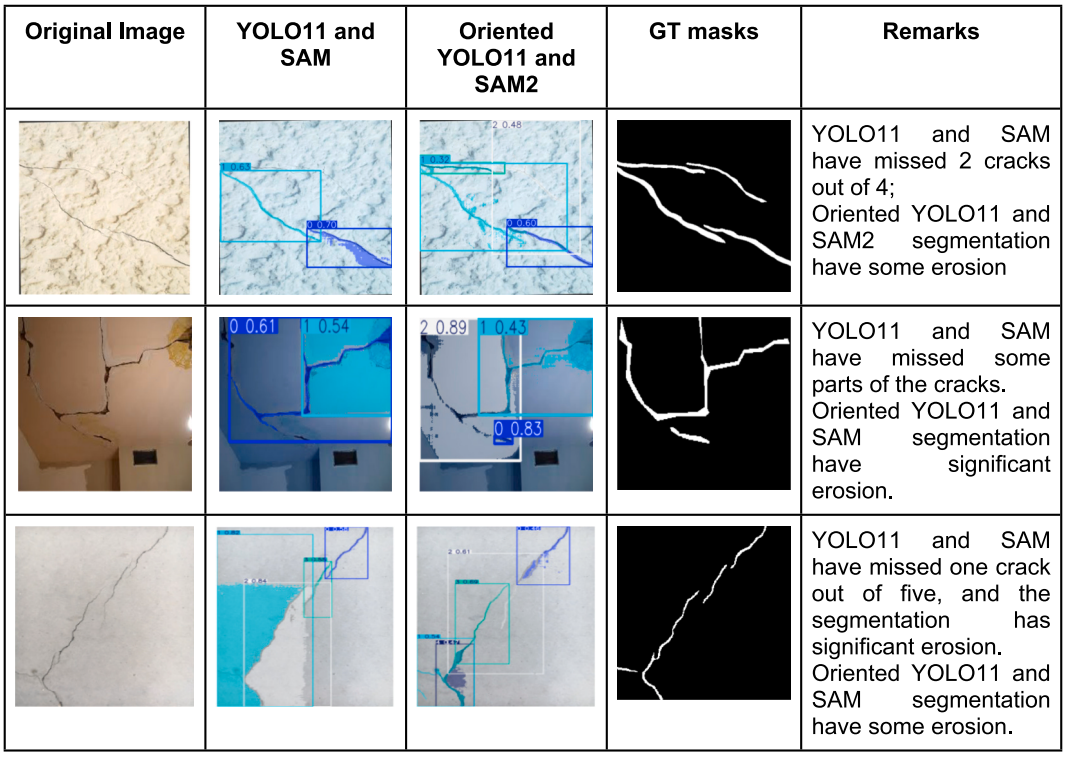


Fig. 16. Combined box and point-based prompt.

Table 3
Different segmentation models details.

Model	YOLO11n-seg	UNet	Mask R-CNN	DEEPLABV3+	SegFormer	YOLO11-SAM
Layers	355	50	450	400	40	355 (object detection) 24 (SAM)
No. of Parameters	2.9 M	31 M	44 M	43 M	3.7 M	2.9 M (object detection) 110 M (SAM)
	Composite loss Box Regression Loss (IoU/GIoU/ DIoU/CIoU), Objectness loss (BCE), Class probability loss (CE/Focal), Mask loss (BCE/Dice).	BCE + Dice combo Binary Cross-Entropy (CE) loss Dice loss	Multi-task loss Classification Loss (CE), Bounding box regression loss (Smooth L1 / L2), Mask loss (per-pixel BCE).	Cross-Entropy/ Weighted CE Loss Dice Loss/ Focal loss	Cross-Entropy Loss Dice Loss/ Focal loss	YOLO11n-Object detection Box Regression Loss (IoU/GIoU/ DIoU/CIoU), Objectness loss (BCE), Class probability loss (CE/Focal), SAM Mask Loss
Hyper parameters	Optimizer: SGD with momentum 0.9 Number of epochs = 20 Batch size = 16 Base Lr = 0.01 Eff. Lr = 0.0025 Weight decay wd = 0.0005	Optimizer: Adam Number of epochs = 20 Batch size = 4 Lr = 0.0001 Weightdecay wd = 0	Optimizer: SGD with momentum 0.9 Number of epochs =20 Batch size: 2 Base Lr = 0.00025 Weight decay wd = 0.0001	Optimizer: Adam Number of epochs = 20 Batch size = 4 Lr = 0.0001 Beta = 0.9 Weightdecay wd = 0	Optimiser: Adam Number of epochs = 20 Batch size = 8 Lr = 0.00005 Weight decay wd = 0.0001	Optimizer: Adam Number of epochs = 20 Batch size = 16 Lr = 0.01 Weight decay wd = 0.0005

For k-means segmentation, the parameters are Number of clusters: 2/3; Maximum iterations: 100; Epsilon (convergence): 0.2; Initialisation method: randomly selected initial cluster centres.

3.2. Dataset annotation

The images have been annotated with rectangular and oriented bounding boxes for object detection and polygons for segmentation, and labelled using LabelMe and Roboflow. The polygon annotations have pixel-perfect precision while being adaptable and flexible over a wide range of real-life scenarios. Thus, these regions enclosed by polygons facilitate the precise identification and quantification of structural defects. The annotated images, which highlight various types of structural defects, serve as the input dataset. These images are divided into training, validation, and testing to ensure the robust evaluation of the proposed segmentation model and other established models.

3.3. Performance evaluation metrics

The evaluation metrics, such as Precision (P), Recall (R), and the Dice

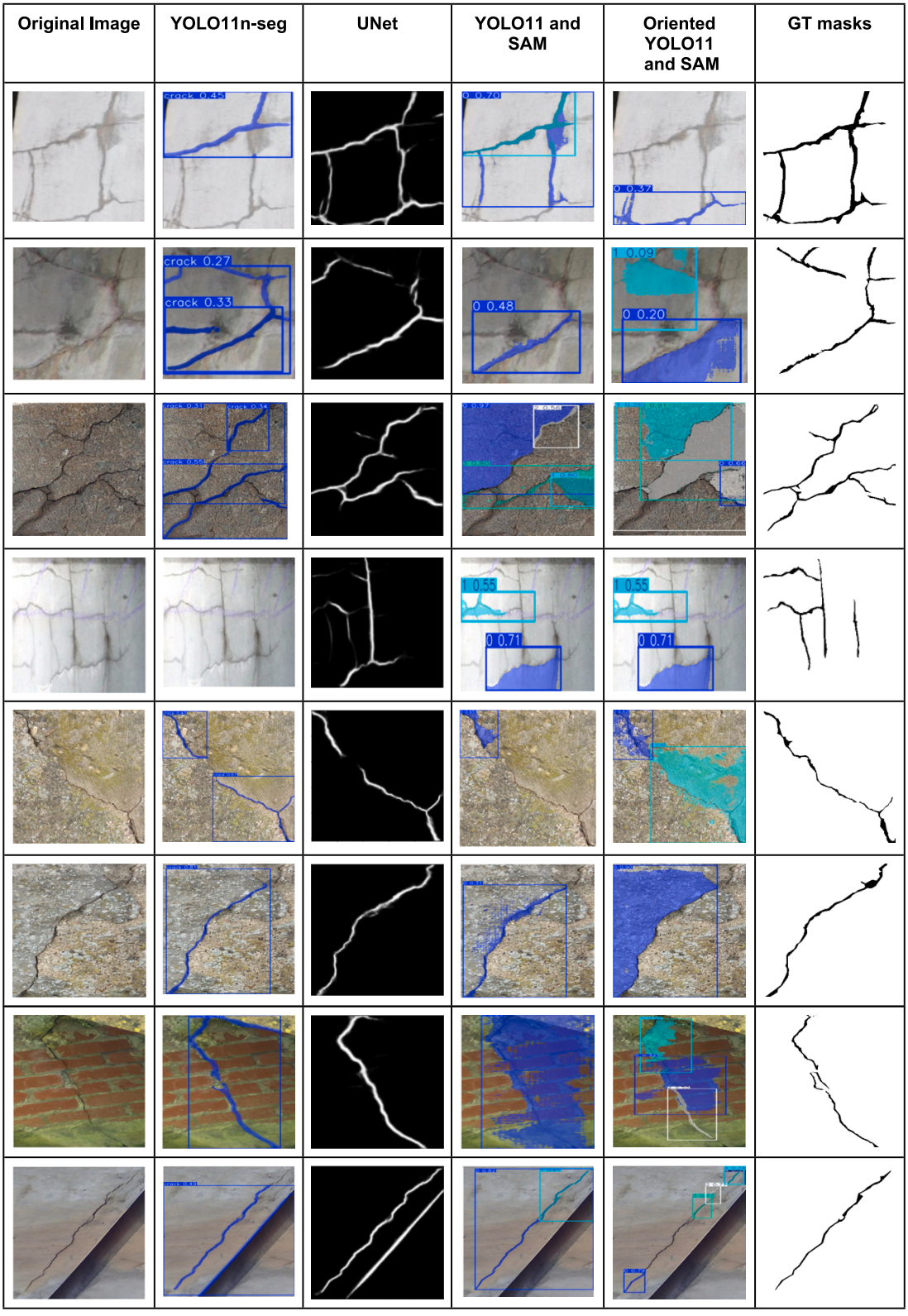


Fig. 17. Performance evaluation of the proposed semi-supervised segmentation model with established segmentation models on the MCDS dataset.

coefficient, are used to assess the performance of the segmentation models. Precision refers to the proportion of correctly identified damage instances among all instances that the system has classified as defective. High precision indicates a low rate of false alarms in damage detection. Recall denotes the proportion of actual damage cases that are correctly identified by the monitoring system. A high recall value implies that the system is effective at detecting most or all of the existing structural damages. Higher values of both P and R indicate better detection capabilities. However, relying solely on Precision and Recall can be misleading, as they do not provide a comprehensive view of model

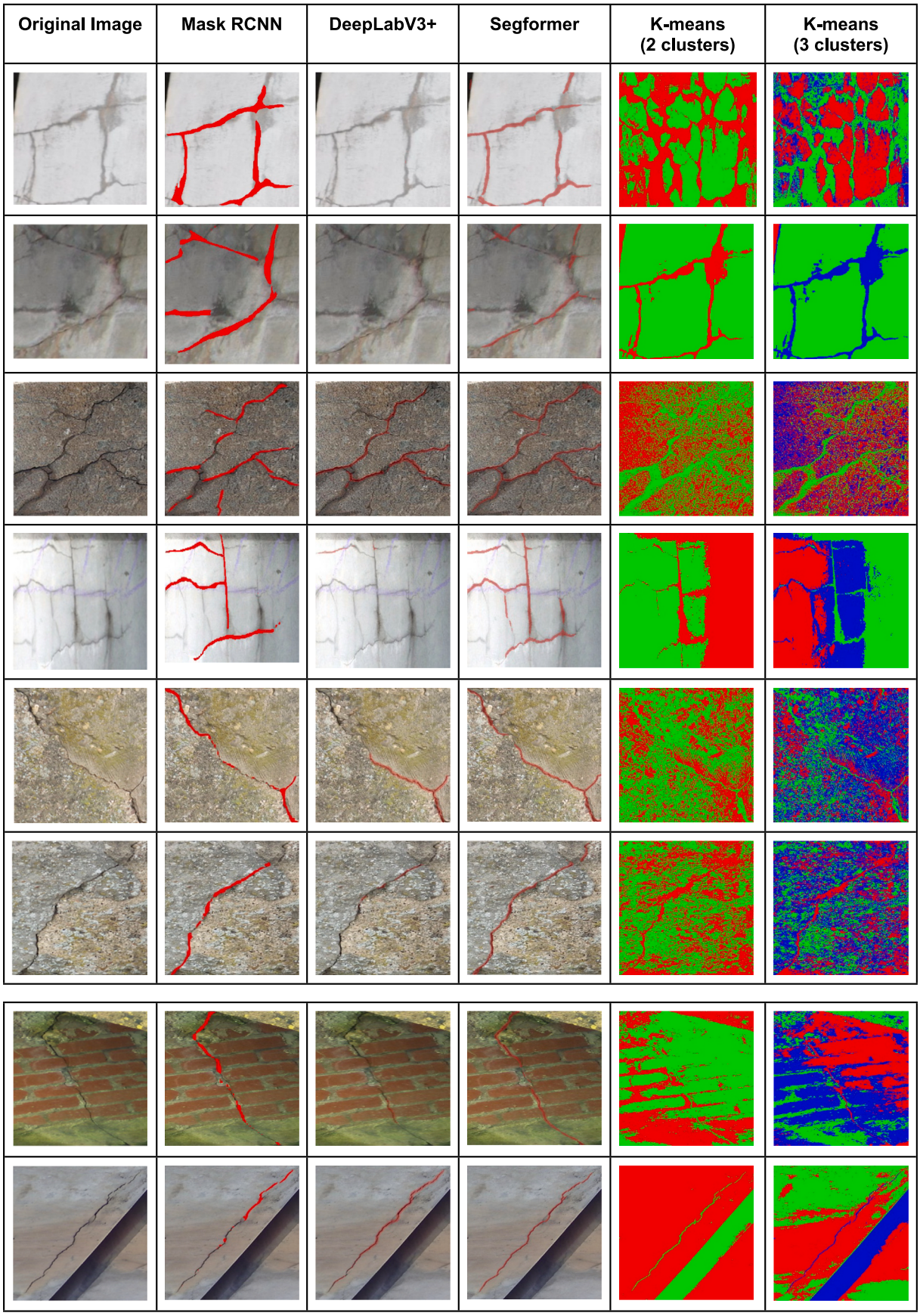


Fig. 17. (*continued*).

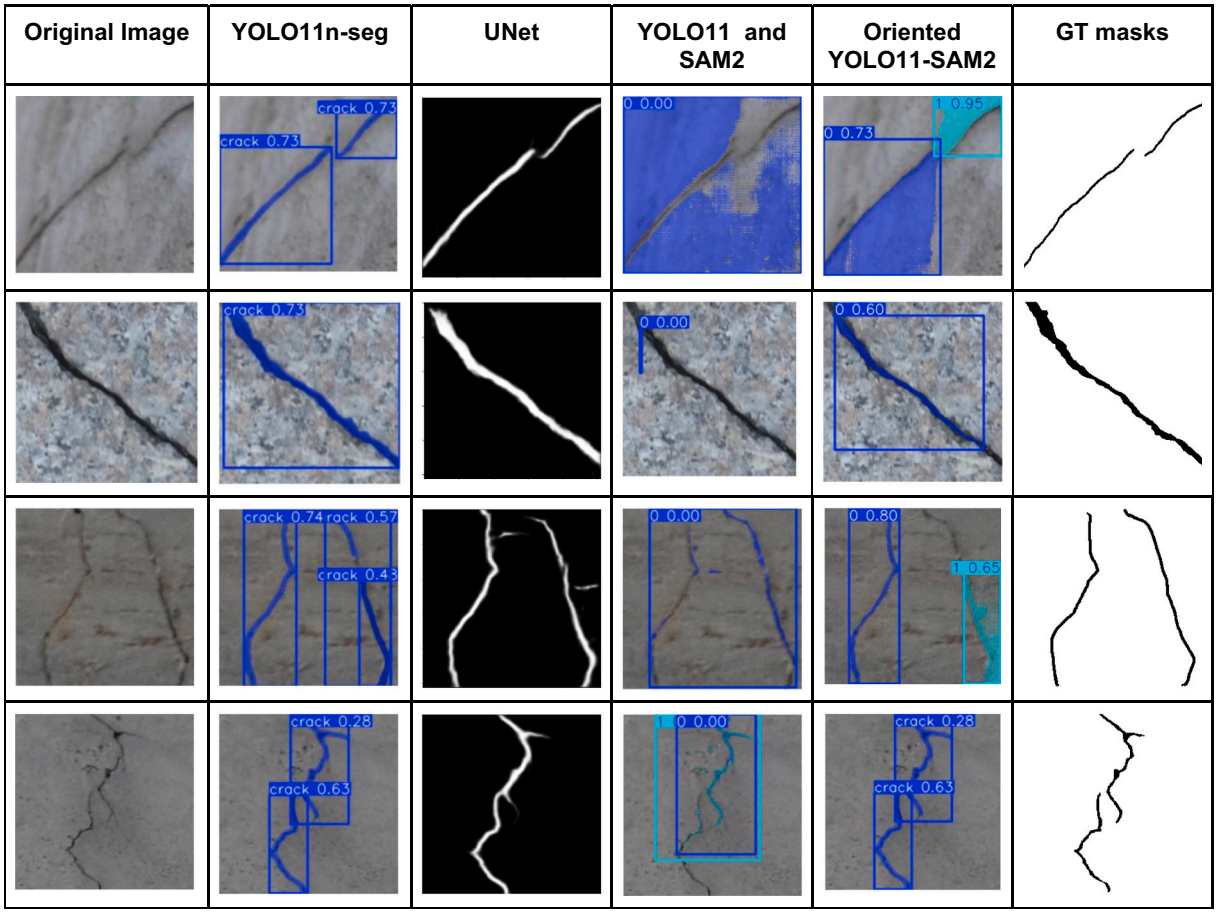


Fig. 18. Performance evaluation of proposed semi-supervised segmentation model with an established segmentation models on BCL - Non-Steel crack dataset.

performance, particularly in cases of class imbalance. The Dice coefficient addresses this challenge by harmonising P and R. The Dice coefficient is particularly useful for evaluating the overlap between predicted and ground truth crack regions, ensuring a more accurate assessment of segmentation effectiveness.

To quantify the uncertainty of these metrics, a 95 % confidence interval can be estimated using the non-parametric bootstrap method. In this approach, multiple bootstrap samples are generated by resampling (with replacement) from the original test set. The metric (e.g., Dice coefficient) is computed for each bootstrap sample, and the 2.5th and 97.5th percentiles of the resulting distribution define the 95 % confidence interval. This interval provides a robust estimate of the range within which the true metric is expected to lie, offering insight into the variability of model performance.

The expressions of P, R, Dice coefficient/Dice Score(DS) and Intersection over Union (IOU) are given below.

$$\text{Precision (P)} = \text{TP}/[\text{TP} + \text{FP}] \tag{1}$$

$$\text{Recall (R)} = \text{TP}/[\text{TP} + \text{FN}] \tag{2}$$

$$\text{Dice Coefficient (Dice Coeff)} = (2 \times \text{P} \times \text{R}) \div (\text{P} + \text{R}) \tag{3}$$

$$\text{Intersection Over Union (IOU)} = \text{TP}/[\text{TP} + \text{FP} + \text{FN}] \tag{4}$$

Where TP, FP, and FN represent True Positive, False Positive, and False Negative, respectively.

4. Proposed semi-supervised segmentation model evaluation on crack datasets (single and cross-dataset training and testing)

The domain of crack detection faces critical challenges due to the absence of large-scale, standardised datasets that encompass diverse surface textures, environmental variations, and crack patterns. The limited scope of existing datasets restricts the ability of models to develop robust features suitable for varied real-world applications. Moreover, inconsistencies in data collection methods and labelling practices hinder reproducibility, making objective comparisons between different approaches difficult. To address these gaps, there is a pressing need for extensive, standardised datasets that represent a wide array of real-world conditions and infrastructure types, coupled with uniform protocols for data collection and annotation. Despite notable progress in the field, issues related to model adaptability, computational performance, the integration of multimodal data, and the practical utilisation of crack analysis results for informed decision-making persist [17].

In view of this, the proposed segmentation models: Hybrid YOLO11 with SAM and Hybrid Oriented YOLO11 with SAM were trained separately on four crack datasets and tested on the corresponding and cross-dataset test subsets to assess generalisation performance. The proposed semi-supervised segmentation model is evaluated on both homogeneous ("within dataset") and heterogeneous ("across dataset") training and testing, and its performance is also compared with several state-of-the-art supervised algorithms.

Cross-dataset testing on unseen datasets is a standard approach to evaluate domain shift generalisation. This cross-dataset testing enables us to assess how the model performs and handles domain shift when there are differences in terms of image resolution, lighting, or defect

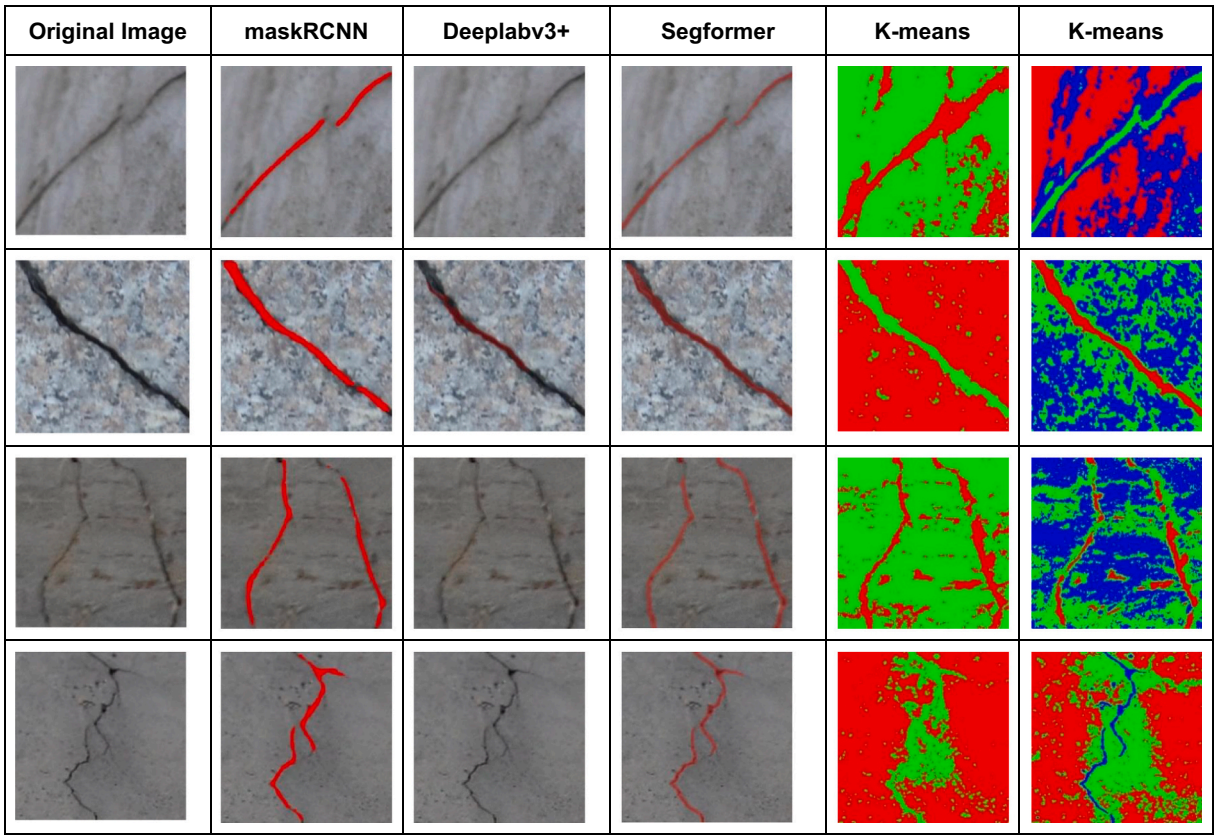


Fig. 18. (*continued*).

patterns. The following training and testing scenarios are considered.

i) Train with BPHDR and test on BPHDR.
ii) Train with BPHDR and test on MCDS.
iii) Train with BPHDR and test on BCL_non steel.
iv) Train with BPHDR and test on BCL_steel.
v) Train with MCDS and test on BCL_non steel.
vi) Train with BHPDR, MCDS, and BCL_steel and test on BCL_non steel.

Although training and testing are conducted on all four crack datasets using the same dataset, only the results corresponding to the BPHDR dataset are presented for brevity. Similarly, while all possible cross-dataset segmentation scenarios were investigated, only selected representative results are shown to avoid redundancy. It should be noted that testing is conducted on completely unseen data.

To address domain shift between training and test data, explicit domain adaptation techniques such as feature alignment, style transfer, or self-supervised learning have shown promise in improving cross-domain generalisation. Although the present study does not incorporate explicit domain adaptation methods, the sixth scenario involves training on multiple diverse datasets and testing on an unseen dataset, which partially mitigates this issue. Transfer learning with multiple datasets of similar defect classes enhances performance and mitigates the issue of domain shift between training and testing data. This strategy enhances the robustness of the proposed model by exposing it to a wide variety of defect characteristics and imaging conditions, thereby improving its ability to generalise across domains.

This section is dedicated to investigating the comparative performance of the proposed segmentation models: (1) Hybrid YOLO11 and SAM, and (2) Hybrid Oriented YOLO11 and SAM. Accordingly, the traditional box-based prompt is employed in the second stage to facilitate unsupervised segmentation.

The four crack datasets considered by the models include different orientations of cracks, such as horizontal, vertical, oriented, irregular/alligator, multiple, connected, and disconnected cracks, as well as cracks with stains, handwritten marks, similar edges, different backgrounds, noisy images, and cracks in various materials.

Figs. 8 to 13 present the input image and the output image with pixel-level segmentation mask predicted by the proposed hybrid YOLO11 and SAM, as well as the hybrid Oriented YOLO11 and SAM model for the training and testing scenarios I to VI, respectively. Table 2 presents the performance evaluation metrics estimated on the testing dataset by the proposed hybrid YOLO11 and SAM, as well as the hybrid Oriented YOLO11 and SAM, for scenarios I to VI. The maximum and average dice scores, along with the 95 % confidence interval obtained for the testing dataset, are presented in Table 2.

The following observations can be made from Figs. 8 to 13 and Table 2.

1. It is evident from Figs. 8(d), 8(e), 10(e), 10(a), and 10(e), which correspond to straight cracks i.e. either horizontal (transverse) or vertical (longitudinal), that both the hybrid YOLO11 and SAM, and the hybrid Oriented YOLO11 and SAM models yield comparable results. However, the proposed Horizontal YOLO11 and SAM model tends to include a substantial amount of background within the bounding boxes, as observed in Fig. 10(e).
2. In the case of inclined cracks, as shown in Figs. 8(a), 8(b), 9(f), 9(g), 9(h), 10(a), 10(b), 11(b), 11(c), and 11(f), the hybrid Oriented YOLO11 and SAM model demonstrates superior performance in generating accurate segmentation masks compared to the hybrid YOLO11 and SAM. Notably, the hybrid YOLO11 and SAM model often fails to capture certain cracks and includes excessive background, as seen in Figs. 8(a) and 10(a).
3. For edge cracks, Fig. 11(c) shows that although both the hybrid YOLO11 and SAM, as well as the hybrid Oriented YOLO11 and SAM

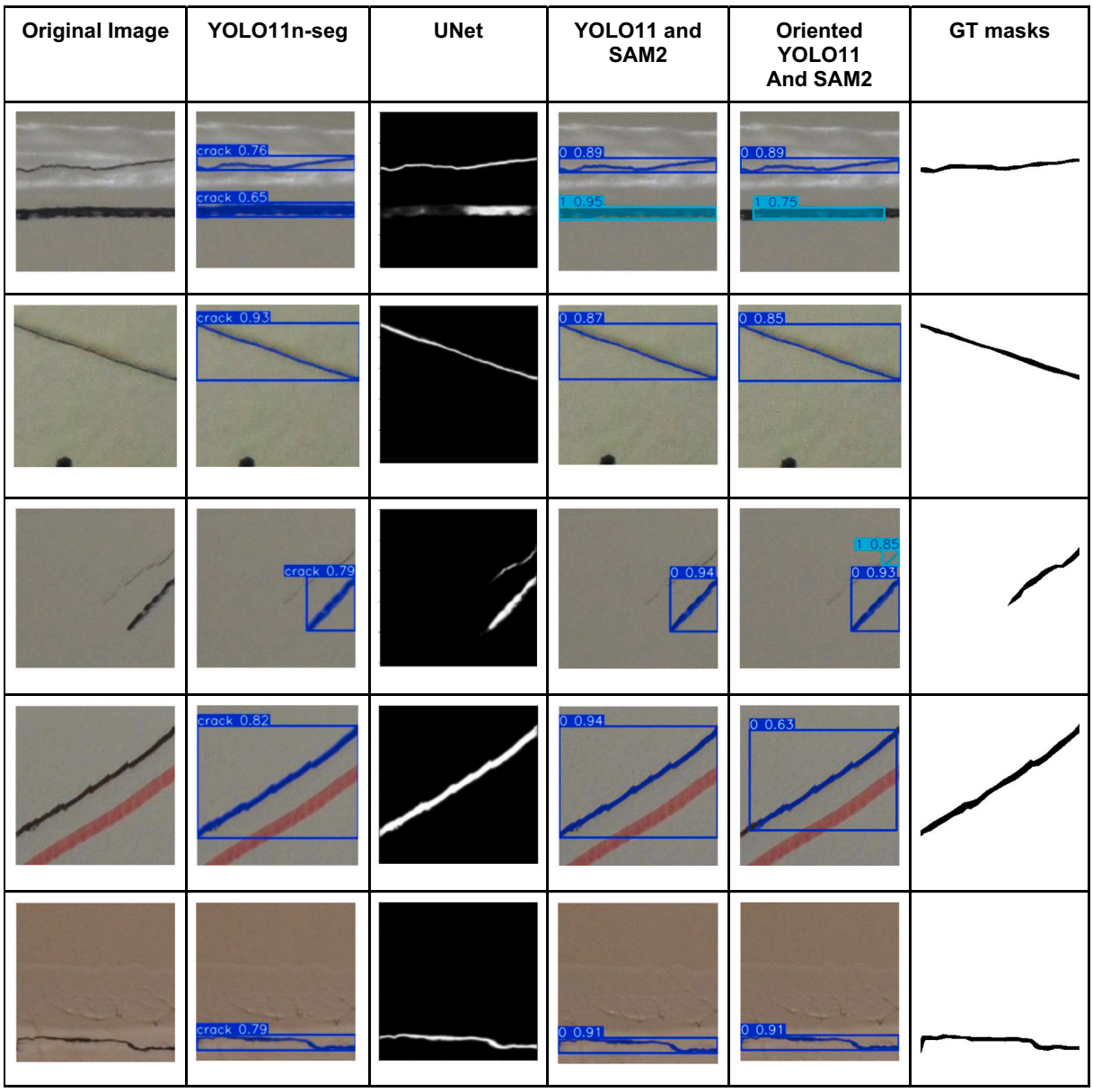


Fig. 19. Performance evaluation of proposed semi-supervised segmentation model with widely used segmentation models on BCL - Steel crack dataset.

models, miss some portions of the cracks, the latter produce segmentation masks that more closely align with the ground truth.

4. Figs. 8(a), 9(e), 9(f), 9(g), 10(b), 10(d), 11(f), and 11(c), which pertain to cracks embedded in complex or noisy backgrounds—such as tiled or masonry surfaces and regions with shadows or spurious holes—indicate that the hybrid Oriented YOLO11 and SAM model outperforms the hybrid YOLO11 and SAM in terms of mask accuracy and background separation.
5. Regarding irregular, alligator, or randomly distributed cracks, as seen in Figs. 8(c) and 9(a) through 9(d), the hybrid Oriented YOLO11 and SAM again shows improved segmentation performance. Nevertheless, the model also exhibits limitations in certain challenging cases, particularly in Figs. 9(a) and 9(d), where the predicted masks deviate significantly from the ground truth.
6. Figs. 11(a) and 11(d), which involve cracks with interfering features such as stains, handwritten marks, or visually similar edges, reveal that both hybrid models struggle to achieve satisfactory segmentation accuracy.
7. Finally, comparisons between Figs. 10(a) and 13 (a), Figs. 10(b), 12 (b) and 13(b), Figs. 10(c) and 13(c) covering inclined, irregular, and multiple cracks demonstrate that both the hybrid YOLO11 and SAM and the hybrid Oriented YOLO11 and SAM models are capable of accurate defect segmentation even when trained and tested on datasets from different sources. Notably, cross-dataset training leads to improved performance, as evidenced by the higher confidence scores observed in Figs. 13(a) to 13 (c).
8. It can be observed from Table 2 that cross-dataset training and testing yield similar performance to that of same-dataset training and testing. In contrast, training and testing on different datasets exhibit poor performance.

5. Proposed semi-supervised segmentation models (1. Hybrid YOLO11 and SAM and 2. Hybrid oriented YOLO11 and SAM) with different types of prompts

Three different prompt types—(i) point-based, (ii) box-based, and

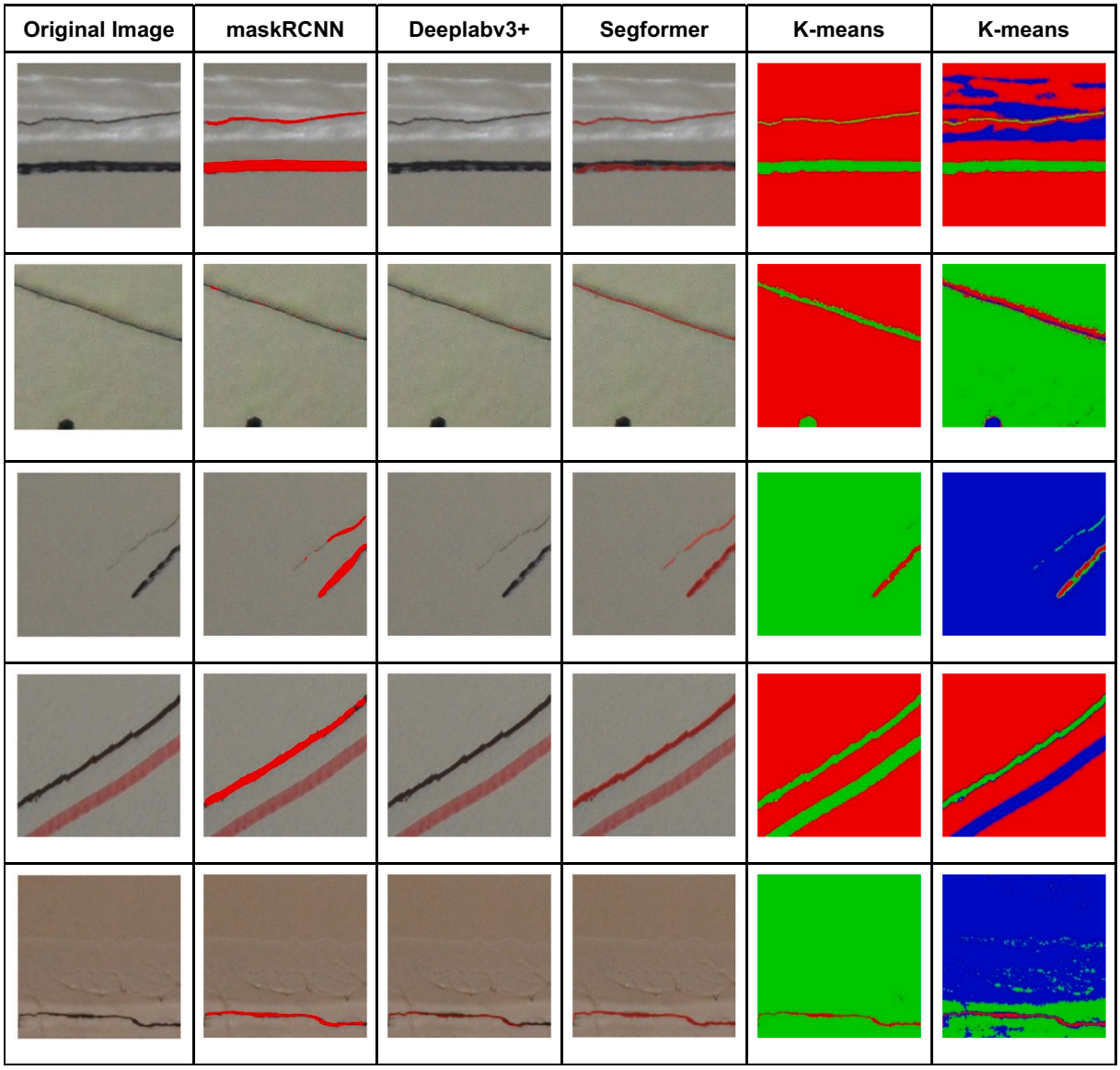


Fig. 19. (*continued*).

(iii) hybrid box-point prompts are systematically evaluated for zero-shot segmentation in combination with both rectangular and oriented YOLO11 models. While box-based prompts are traditionally used, point-based prompts offer greater specificity, particularly when multiple objects exist within a single bounding box, thereby improving target identification and reducing the inclusion of irrelevant background. The hybrid prompt approach, which combines box and point inputs, was introduced to better capture the complexity of crack propagation, particularly in cases with irregular orientations. Figs. 14, 15 and 16 present the sample results of the comparison of different types of prompts, namely box, point, and combined box and point-based prompts, along with the YOLO11, SAM, and Oriented YOLO11 and SAM algorithms towards segmentation. The methodology was evaluated using different kinds of images containing branched cracks, multiple cracks, noisy textures, and other challenging scenarios.

The following observations can be made from Figs. 14, 15 and 16.

1. Fig. 15 indicates that both the proposed hybrid YOLO11 and SAM, as well as the hybrid Oriented YOLO11 and SAM models, fail to detect certain cracks when used with the point-based prompt.
2. As shown in Fig. 14, the use of the box-based prompt yields improved segmentation performance for both models across all crack datasets compared to other prompt types.
3. Furthermore, Fig. 14 highlights that the hybrid Oriented YOLO11 and SAM model, when used with the box-based prompt, yields segmentation masks that more closely align with the ground truth than those generated by the hybrid YOLO11 and SAM model under the same conditions.
4. In Fig. 16, both models, when combined with the box- and point-based prompts, demonstrate accurate segmentation of cracks in challenging scenarios involving multiple fine cracks and backgrounds with similar textures. However, regional erosion of the segmentation masks is also observed.
5. Collectively, Figs. 14, 15 and 16 suggest that the hybrid YOLO11 and SAM model shows enhanced segmentation performance when using the combined box- and point-based prompt compared to its performance with individual prompts.
6. Notably, the hybrid Oriented YOLO11 and SAM model, when applied with the combined box-based prompt, delivers the best segmentation results, outperforming not only its own performance with other prompts but also that of the hybrid YOLO11 and SAM model under

Table 4
Performance evaluation of different models on the crack dataset.

Models	Precision	Recall	mIOU	AUPRC	Dice Coefficient (mean & CI)
YOLO11n-seg	0.759	0.659	0.578	0.709	0.705 95 % CI:[0.665 0.766]
UNet	0.802	0.677	0.505	0.742	0.735 95 % CI:[0.616 0.746]
DeepLabv3+	0.512	0.502	0.354	0.507	0.517 95 % CI:[0.485 0.619]
Mask R-CNN	0.678	0.552	0.458	0.615	0.621 95 %CI:[0.573 0.687]
Segformer	0.723	0.762	0.589	0.743	0.742 95 % CI:[0.718 0.864]
YOLO11 and SAM	0.854	0.703	0.601	0.779	0.771 95 % CI:[0.734 0.828]
Oriented YOLO11 and SAM	0.893	0.759	0.690	0.826	0.820 95 % CI:[0.793 0.867]

any prompt configuration. Based on these findings, the study concludes that the hybrid Oriented YOLO11 and SAM model with the box-based prompt offers the most effective approach for crack segmentation.

6. Comparison of the proposed semi-supervised model with supervised segmentation models for the crack dataset

To evaluate the performance of the proposed hybrid YOLO11 and SAM, the hybrid Oriented YOLO11 and SAM model in segmenting the defects at the pixel level over the established direct supervised models, such as YOLO11-Seg, UNet, Mask R-CNN, DeepLabV3+, SegFormer, and unsupervised K-means segmentation model, the cross-dataset training and testing scenarios II to IV are considered. The loss functions, hyperparameters, and number of parameters of different segmentation models are presented in Table 3.

6.1. Influence of different hyperparameter tuning of different models towards fairer comparison

Matching input resolution and data augmentation protocols is crucial for meaningful benchmarking, as differences here can overshadow genuine architectural advantages or weaknesses.

Different models often require distinct hyperparameter settings due to variations in their architectures and training dynamics. Using the same optimiser or learning rate across models can lead to suboptimal performance and biased comparisons. Optimisers and learning rates are typically fine-tuned to minimise training and validation loss, balancing convergence speed and stability. The learning rate controls the step size for parameter updates: if it is too high, the model may oscillate or diverge; if it is too low, training slows or becomes trapped in suboptimal minima. Inappropriate optimisers or learning rates can produce noisy gradients or irregular loss curves, whereas careful tuning smooths training and prevents breakdowns [45–50]. Transformer-based models, such as SegFormer, typically require lower learning rates and benefit from the AdamW optimiser due to their sensitivity to gradient noise and the scale of parameter updates. In contrast, CNN-based architectures, such as DeepLabV3+, Mask R-CNN, and U-Net, perform robustly with SGD variants or Adam, commonly tolerating higher learning rates. Hybrid models, such as YOLO11, combined with SAM often benefit from moderate to high learning rates [48–53].

Batch size also affects training stability, convergence speed, and generalisation by influencing the noise in gradient estimates and the update frequency. Larger batches produce more stable gradients and faster per-epoch training, but may increase the risk of overfitting. In contrast, smaller batches introduce gradient noise that can help escape local minima, but potentially cause instability. Maintaining a consistent batch size helps isolate architectural differences, while joint tuning of batch size and learning rate ensures comparable training dynamics across models. Empirically, increasing the batch size often allows for proportional scaling of the learning rate (linear scaling rule) to maintain similar gradient behaviour; failing to adjust the learning rate may degrade performance. The "linear scaling rule"—increasing the learning rate as batch size increases—has strong empirical and theoretical support, allowing models to benefit from larger batches without degraded training dynamics [45–50]. By jointly optimising batch size and learning rate, convergence stability, training speed, and overall performance can be aligned across models, allowing for fair and efficient comparisons.

Figs. 17 to 19 present the input image, GT mask and the output image with pixel-level segmentation mask predicted by the proposed hybrid YOLO11 and SAM, the hybrid Oriented YOLO11 and SAM model and the several widely used direct supervised segmentation models such as YOLO11n-Seg, UNet, DeepLabV3+, Mask R-CNN, SegFormer and an unsupervised k-means (2 and 3 clusters) segmentation model for the training and testing scenarios II to IV, respectively. The ground truth (GT) masks are also presented in Figs. 17 to 19 to understand the model performance and for comparison. Table 4 presents the performance evaluation metrics on the testing images by the proposed models, as well as several widely used segmentation models.

Fig. 17 illustrates performance on cracks with diverse textures and complex backgrounds. Figs. 18–19 show results for low-contrast images, stains, and handwritten markings. Fig. 20 presents results for images affected by noise, varied lighting, contrast variations, and occlusions.

The following observations can be made from Figs. 17 to 20 and Table 4.

1. Fig. 17 shows that UNet and SegFormer perform better in segmenting various types of cracks, including straight cracks (horizontal or vertical), irregular cracks, alligator cracks, connected and disconnected cracks, and cracks in complex or noisy backgrounds. UNet is effective for semantic segmentation but does not support instance segmentation. The proposed hybrid YOLO11 and SAM and the hybrid Oriented YOLO11 and SAM models were unable to accurately detect multiple cracks and alligator cracks in complex backgrounds, indicating a need for further improvement. The challenges faced by the YOLO11 and SAM model stem from several factors: (i) irregular and alligator cracks often have complex, less distinct patterns that are difficult to capture, (ii) very fine cracks may contain too few pixels for reliable detection by YOLO11 and precise mask generation by SAM, and (iii) SAM generates smooth region-level masks, which may not capture the intricate boundaries of alligator cracks.
2. Figs. 17 to 19 indicate that YOLOv11n-Seg and Mask R-CNN generally perform well across different crack types; however, they occasionally fail to detect irregular and alligator cracks in images with noisy or complex backgrounds.
3. Figs. 18 and 19 demonstrate that DeepLabv3+ underperforms compared to other models in segmenting cracks. The proposed hybrid YOLO11 and SAM and hybrid Oriented YOLO11 and SAM achieve performance levels comparable to those of direct supervised models, such as UNet, SegFormer, YOLO11n-Seg, and Mask R-CNN.
4. Figs. 17 and 18 show that although K-means segmentation can identify irregular, alligator, and inclined cracks, it often produces fragmented shapes, broken boundaries, noise, and small artefacts that do not correspond to real cracks when applied to images with background noise, varied lighting, and complex textures. Morphological operations (e.g., erosion, dilation, opening, and closing) are therefore critical post-processing steps in K-means segmentation, as they refine, clean, and enhance the segmented regions, correcting

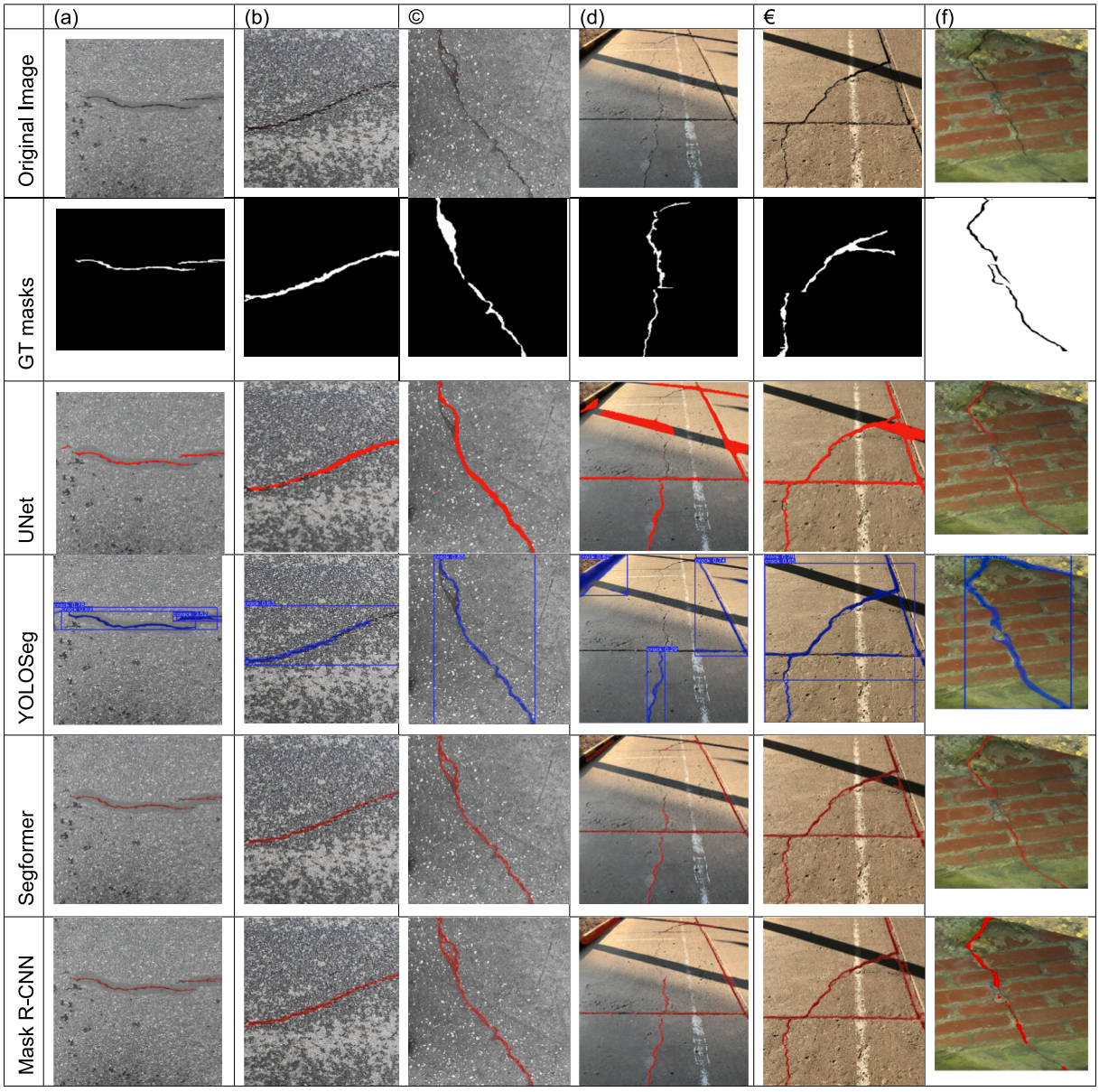


Fig. 20. Performance evaluation of proposed semi-supervised segmentation model with established segmentation models on crack images with background noise, varied lighting and occlusions.

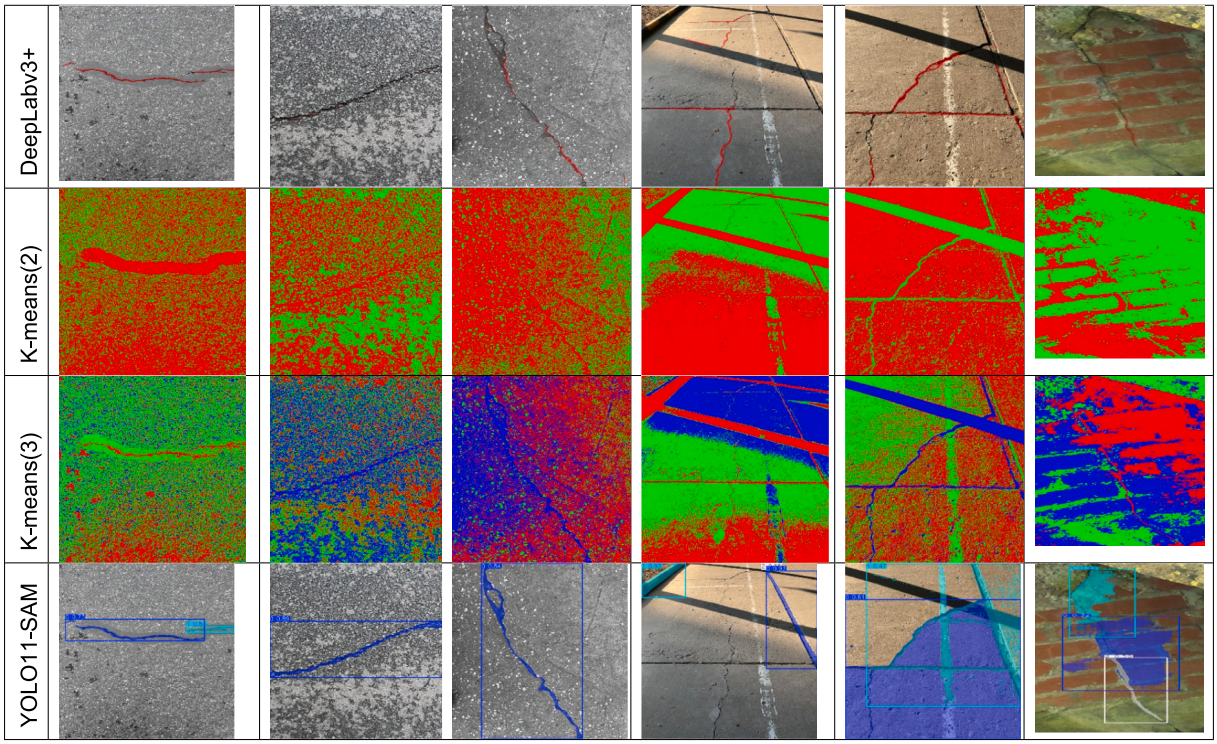


Fig. 20. (*continued*).

imperfections caused by noise, intensity inhomogeneity, and spatial ambiguity when clustering is based solely on pixel values or colours.

5. Fig. 19 illustrates that K-means segmentation can identify cracks in the BCL_steel dataset when using 2 or 3 clusters. However, it also misclassifies handwritten annotations, stains, and artificial artefacts as cracks.
6. It can be observed from Table 4 that the proposed semi-supervised models, namely hybrid YOLO11 and SAM, as well as hybrid Oriented YOLO11 and SAM, achieve similar performance in terms of accuracy and Dice score to that of the direct supervised models.
7. All direct supervised segmentation models, i.e., UNet, DeepLabv3+, Mask R-CNN, Segformer and YOLO11n-Seg, require pixel-level annotation, which is a labour-intensive task, and the style of annotation also affects the number of masks generated. However, for the proposed semi-supervised segmentation algorithms, the accuracy is less influenced by pixel-level annotations and can still perform effectively with coarser annotations.
8. The proposed semi-supervised model does not require a pixel-wise annotated dataset, but only a region-level annotation.The proposed model works for any defect without requiring explicit training with pixel-annotated images, as it performs zero-shot segmentation using the SAM. A comparison between region-level annotation and pixel-level annotation reveals that pixel-level annotation is significantly more complex, especially for smaller or thinner cracks. Additionally, the choice of annotation (whether area-based or fine-grained, polygon simplifications, or noisy annotations) can influence the model's performance. Region-level annotations are less impacted by the size of the crack and show greater robustness to variations in user annotation style. Region-level annotation is more robust to variations in crack size and user annotation styles, offering greater stability across diverse datasets.

Fig. 20 presents the performance evaluation of the proposed semi-supervised segmentation model compared with established segmentation models on crack images under challenging conditions, including background noise, varied lighting, and occlusions. From Figs. 20(a) to 20(c), it can be observed that the proposed hybrid YOLO11 and SAM model, along with UNet, Mask R-CNN, YOLO-Seg, and SegFormer, successfully identify inclined and irregular cracks despite the presence of background noise. However, Figs. 20(d) to 20(e) show that the proposed hybrid YOLO11 and SAM model fails to detect certain cracks. This limitation can be attributed to the fact that very fine or thin cracks may consist of too few pixels for reliable detection by YOLO11 and for precise mask generation by SAM. Since SAM produces smooth masks from region-level prompts, it may not accurately delineate intricate boundaries in cases of extremely thin cracks.

In addition, complex or noisy backgrounds—such as textures, shadows, stains, or paint variations—can introduce visual clutter and false cues that resemble crack features, thereby confusing the model's attention and leading to missed or incomplete segmentation. The performance of YOLO11 and SAM also depends heavily on the diversity and quality of the training data; limited diversity or coarse annotations may restrict the model's ability to distinguish cracks from cluttered backgrounds. Although the C2PSA module helps mitigate occlusion challenges through cross-scale feature fusion and attention mechanisms, thereby maintaining consistent masks for partially hidden or overlapping defects, the zero-shot segmentation capability of SAM remains less effective for minute or heavily occluded crack structures that require precise pixel-level details. Furthermore, certain loss functions and attention mechanisms may not fully regularise or suppress background noise, further reducing segmentation accuracy in complex cases.

These factors collectively explain why cracks with irregular shapes, specific orientations, smaller sizes, or highly cluttered backgrounds are more challenging for the model to segment accurately. They also highlight potential avenues for improvement, including the incorporation of more diverse datasets, the design of enhanced loss functions, and the introduction of background suppression modules.

Finally, Fig. 20(f) illustrates that the proposed hybrid YOLO11 and SAM model, along with UNet, Mask R-CNN, YOLO-Seg, and SegFormer, can accurately identify cracks even in the presence of occlusions, such as those commonly found in masonry structures. In contrast, K-means segmentation identifies irregular, alligator, and inclined cracks. Still, it

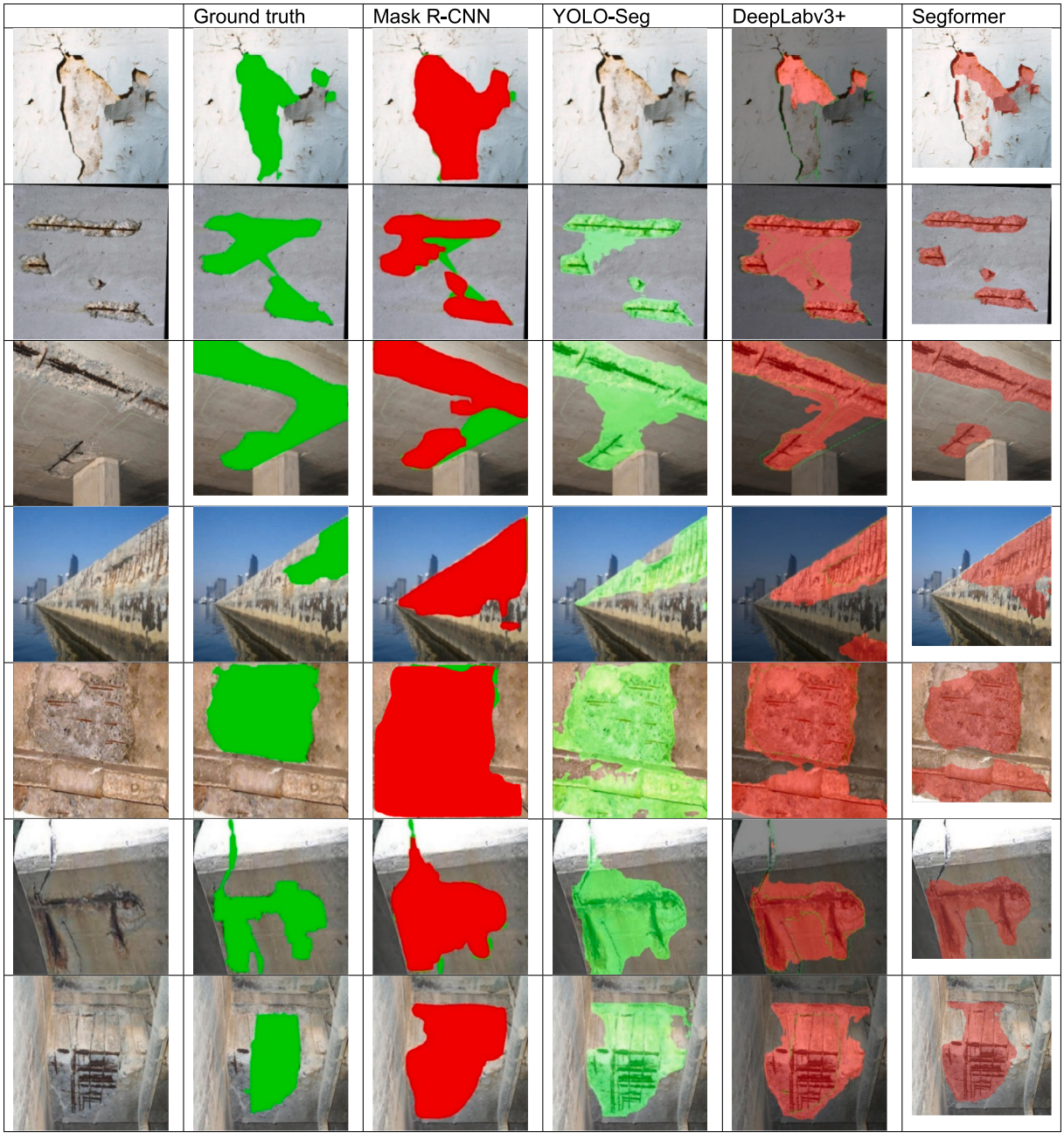


Fig. 21. Performance evaluation of proposed semi-supervised segmentation model with established segmentation models on spalling dataset.

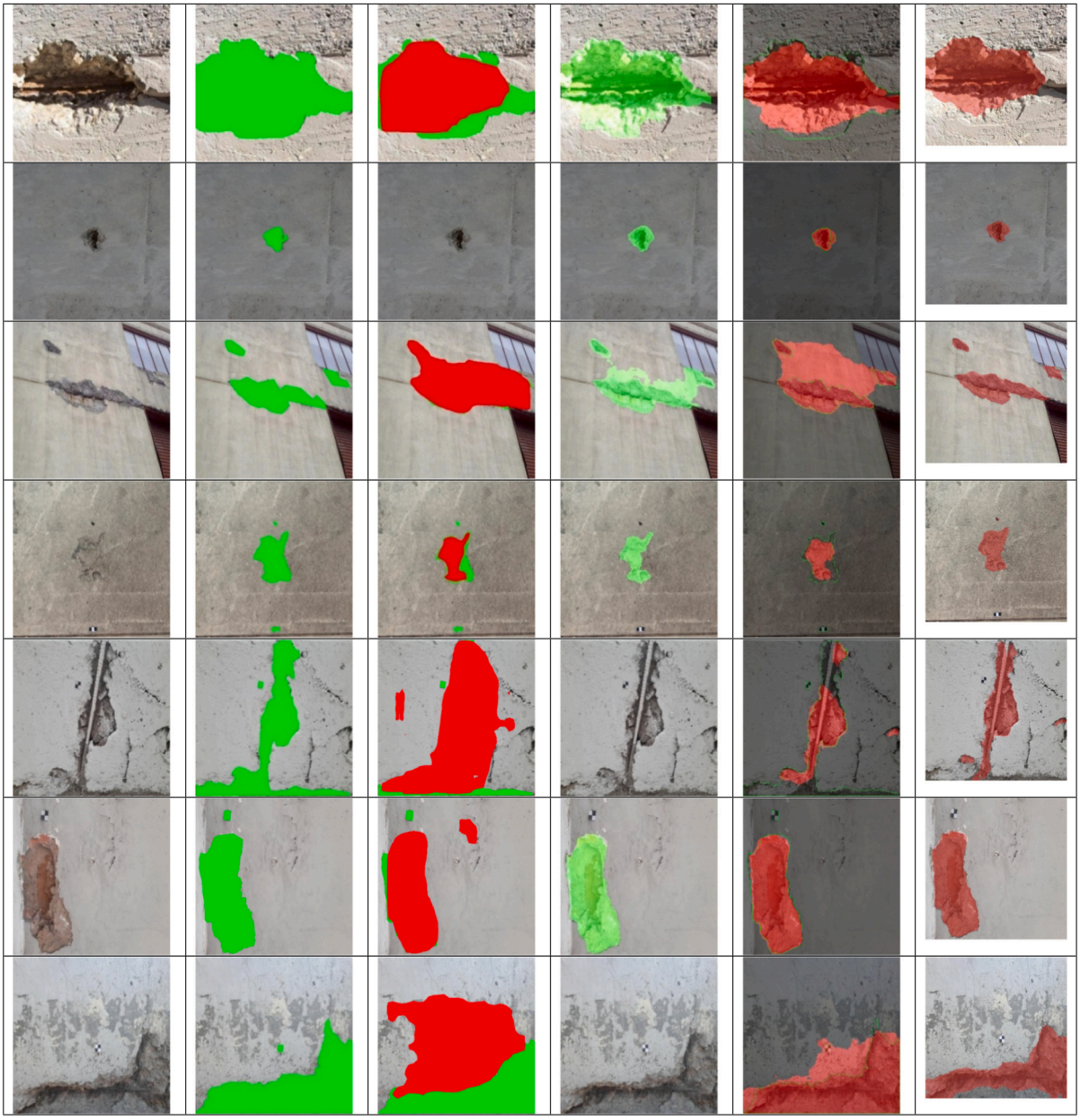


Fig. 21. (*continued*).

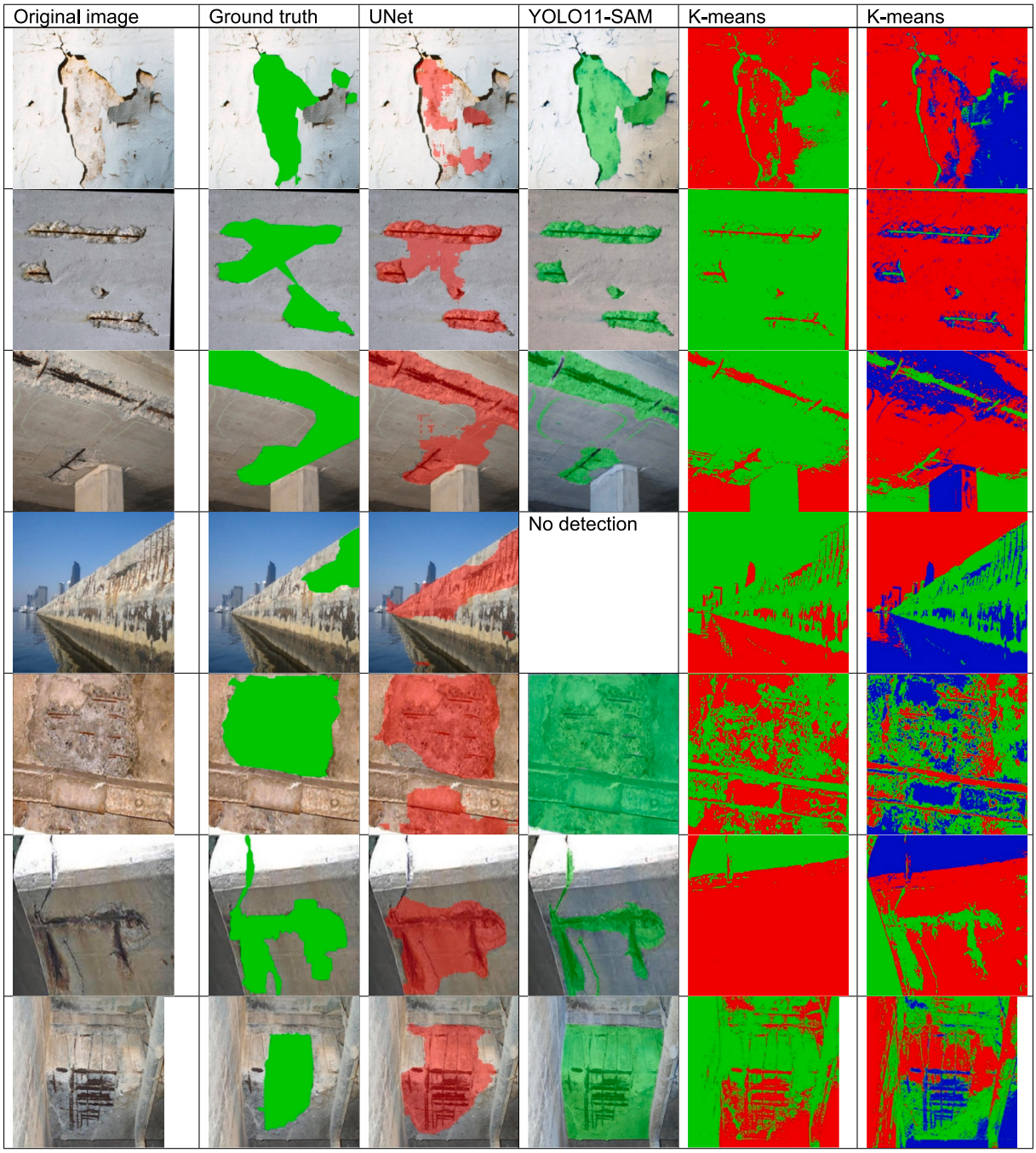


Fig. 21. (*continued*).

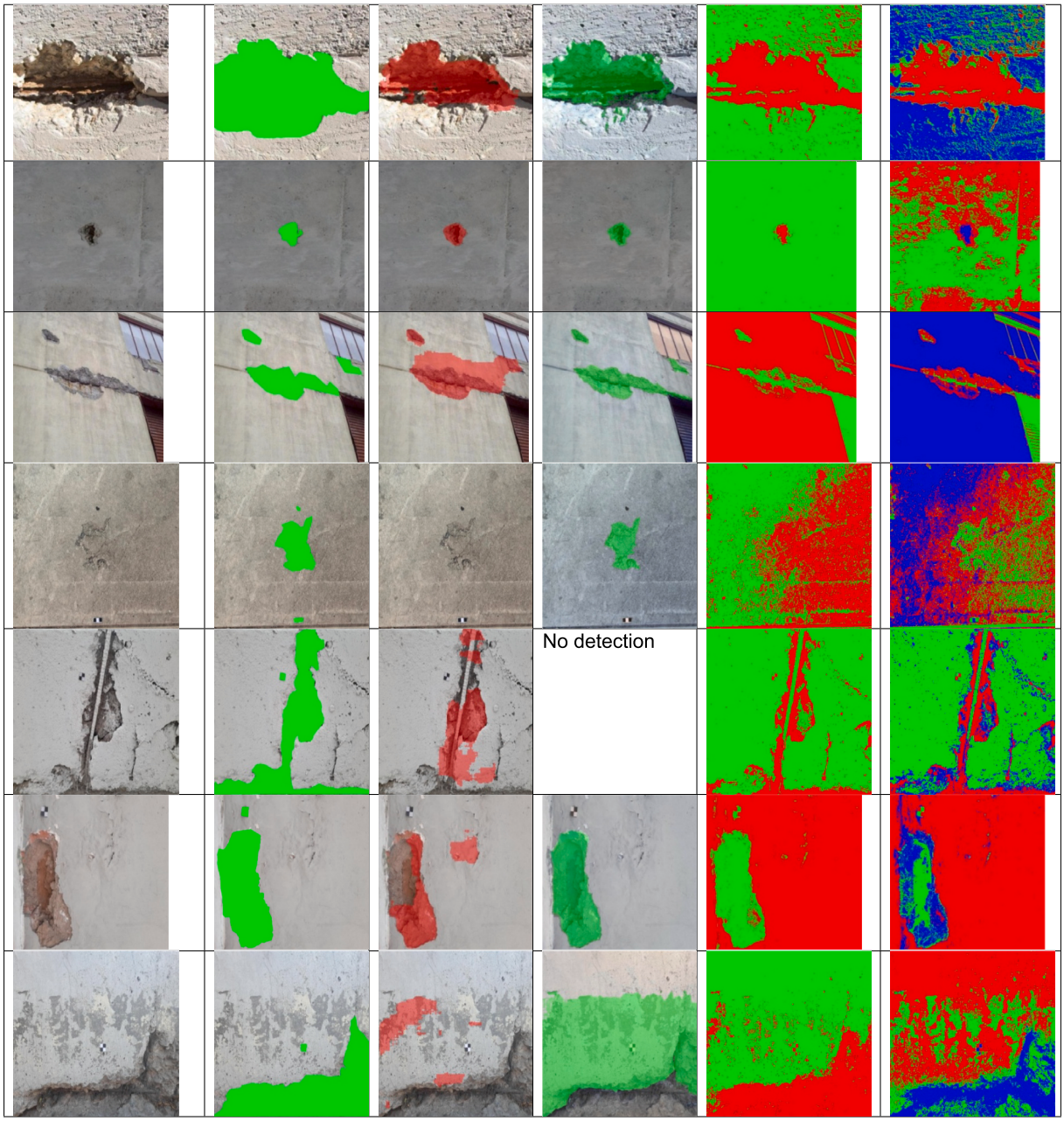


Fig. 21. (*continued*).

Table 5
Performance Evaluation of different models on the spalling dataset.

Models	Precision	Recall	mIoU	AUPRC	Dice Coefficient (mean & CI)
YOLO11n-seg	0.815	0.912	0.667	0.862	0.793 95 % CI: [0.66 0.85]
UNet	0.7538	0.791	0.602	0.775	0.735 95 % CI: [0.63 0.85]
DeepLabv3+	0.756	0.874	0.646	0.818	0.771 95 % CI: [0.706 0.834]
MaskRCNN	0.694	0.905	0.626	0.801	0.756 95 % CI:[0.686 0.826]
SegFormer	0.851	0.789	0.703	0.837	0.808 95 % CI [0.652 0.873]
Proposed YOLO11 and SAM	0.848	0.833	0.693	0.848	0.803 95 % CI [0.652 0.87]

Table 6
Performance evaluation of different models on the corrosion dataset.

Models	Precision	Recall	mIoU	AUPRC	Dice Coefficient
YOLO11n-seg	0.697	0.554	0.419	0.626	0.617 95 % CI: [0.537 0.717]
UNet	0.602	0.360	0.421	0.481	0.571 95 % CI: [0.487 0.673]
DeepLabv3+	0.653	0.422	0.422	0.538	0.592 95 % CI: [0.489 0.695]
MaskRCNN	0.519	0.428	0.384	0.474	0.519 95 % CI: [0.375 0.663]
SegFormer	0.678	0.592	0.444	0.635	0.631 95 % CI: [0.537 0.725]
Proposed YOLO11 and SAM	0.702	0.534	0.438	0.618	0.606 95 % CI: [0.519 0.713]

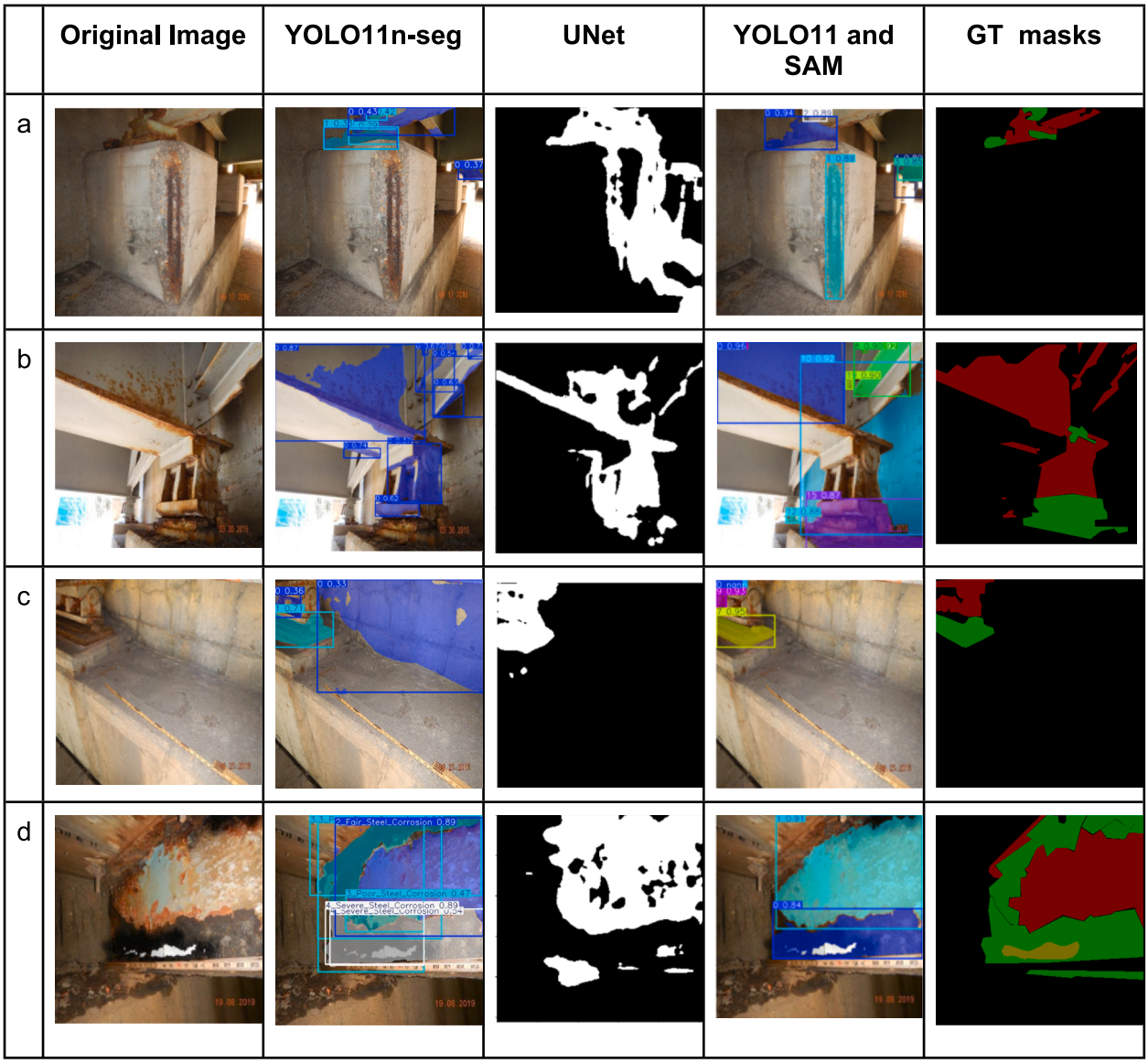


Fig. 22. Performance evaluation of the proposed semi-supervised segmentation model with an established segmentation models on metallic corrosion dataset.

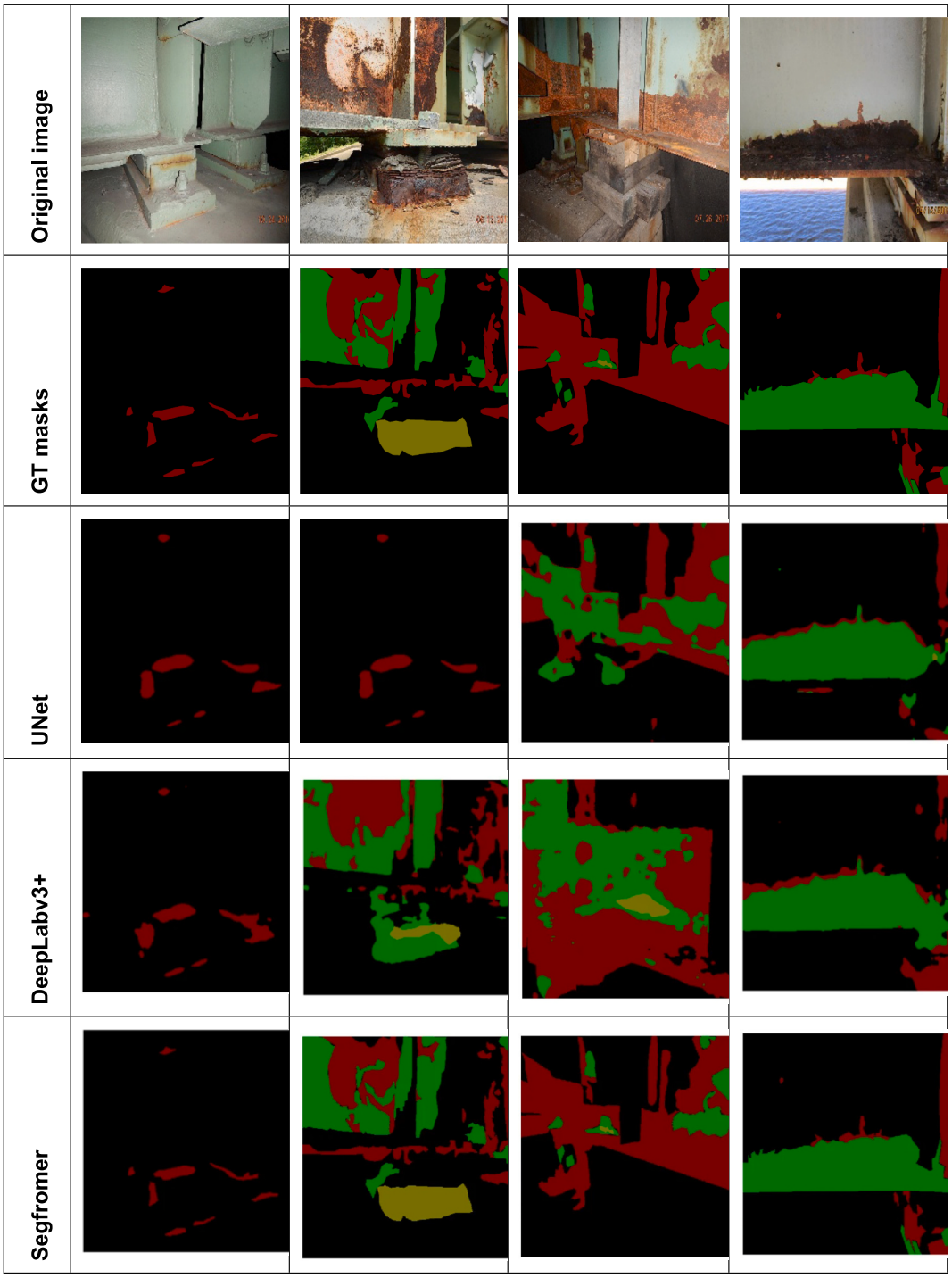


Fig. 23. Performance evaluation of the proposed semi-supervised segmentation model with established supervised segmentation models on metallic corrosion dataset.

often produces fragmented boundaries, noise, and minor artefacts that do not correspond to actual cracks when applied to images with background noise, varied lighting, and complex textures. To address this, morphological operations (e.g., erosion, dilation, opening, and closing) are essential post-processing steps in K-means segmentation, as they refine and enhance the segmented regions, correcting imperfections caused by noise, intensity inhomogeneity, and spatial ambiguity inherent to clustering based solely on pixel values or color.

7. Performance evaluation of the proposed semi-supervised model with supervised segmentation models for the spalling dataset

Besides evaluating the proposed semi-supervised segmentation model on crack datasets, its performance on spalling defect-type segmentation is also investigated. Unlike earlier studies, investigations into spalling defects are limited to those involving only the proposed hybrid YOLO11 and SAM. Spalling defects often manifest as irregular,

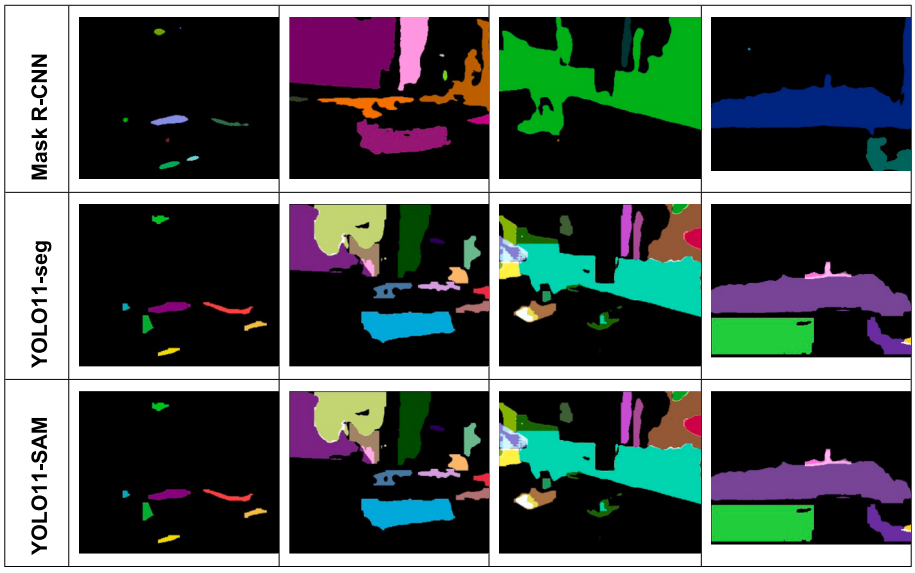


Fig. 23. (*continued*).

polygonal, or amorphous patterns, such as patches or uneven spreads, rather than having a distinct linear or oriented characteristic like cracks.

Fig. 21 presents the input image, GT mask and the output image with pixel-level segmentation mask predicted by the proposed hybrid YOLO11 and SAM, and several widely used direct supervised segmentation models such as YOLO11n-Seg, UNet, DeepLabV3+, Mask R-CNN, SegFormer and an unsupervised k-means (2 and 3 clusters) segmentation model for the spalling dataset. Table 5 presents the performance evaluation metrics on the testing images by the proposed models, as well as several widely used segmentation models on the spalling dataset. The following observations can be made from Fig. 21 and Table 5.

1. It can be observed from Fig. 21 (a) that for coating delamination leading to spalling, both the proposed YOLO11-SAM and Mask R-CNN achieve accurate segmentation, whereas the other models fail to capture the defect reliably.
2. It can be observed from Figs. 21(b) and 21(c) that for elongated or multiple spalling regions with exposed reinforcement, both the proposed YOLO11-SAM, Segformer, and K-means (2 clusters) provide precise segmentation. In contrast, other models tend to over-segment the defects.
3. It can be observed from Fig. 21(d) that only the SegFormer model accurately identifies the spalling.
4. It can be observed from Fig. 21(e) that all the models over-segment in the presence of similar spalling-like patches.
5. It can be observed from Figs. 21(f) to 21(k) that the proposed YOLO11-SAM performs better in identifying thinner edge spalling due to exposed reinforcement.
6. It can be observed from Figs. 21 (l) to 21 (n) that the proposed YOLO11-SAM faces limitations in handling extreme depth or boundary defects. The models DeepLabv3+ and Segformer were able to identify it.
7. It can be observed from Table 5 that the proposed YOLO11-and-SAM model achieves the best balance of high precision, recall, and Dice coefficient, outperforming other state-of-the-art segmentation models on the spalling dataset.

8. Proposed semi-supervised segmentation model evaluation on metallic corrosion dataset

Besides evaluating the proposed semi-supervised segmentation model on crack and spalling datasets, its performance on corrosion defect-type segmentation is also considered, as discussed in earlier sections. Figs. 22 and 23 present the input image, GT mask and the output image with pixel-level segmentation mask predicted by the proposed hybrid YOLO11 and SAM, and several widely used direct supervised segmentation models such as YOLO11n-Seg, UNet, DeepLabV3+, Mask R-CNN, and SegFormer for the corrosion dataset.

Table 6 presents the performance evaluation metrics on the testing images by the proposed models, as well as several widely used segmentation models. K-means clustering was evaluated for unsupervised segmentation of metallic corrosion; however, it consistently produced unsatisfactory results. Corrosion regions exhibit high intra-class variability in color (ranging from orange to dark brown and black), diffuse boundaries, and heterogeneous textures, often resembling background features such as paint, shadows, or stains. Since K-means relies only on pixel intensity and color similarity, it either fragments the surface into irrelevant clusters or merges corroded areas with non-damaged regions. These limitations underscore the inadequacy of traditional clustering-based methods for corrosion detection in real-world settings and highlight the need for advanced, feature-aware frameworks. In this work, a hybrid approach combining YOLO11 (and its oriented variant) with the Segment Anything Model (SAM) is employed, enabling context-driven, robust segmentation of corrosion without requiring dense manual annotations.

The following observations can be made from Figs. 22–23 and Table 6.

1. It can be observed from Fig. 22 that the proposed semi-supervised YOLO11 and SAM could predict segmentation masks for the corrosion defect. However, it misses some finer/highly localized corrosion masks/overlaps two or more regions with the regions of no corrosion, in case of multiple instances of corrosion, when compared to that of UNet and YOLO11n-Seg algorithms. This can be explained by the fact that very small defects may contain too few pixels for the network to generate distinct feature maps. At the same time, region-level labelling provides coarse supervision that can blur boundaries between closely spaced or overlapping defects. Corrosion also exhibits high intra-class variability in color (ranging from orange to dark brown or black), diffuse boundaries, and heterogeneous textures that often resemble background features such as paint, shadows, or stains, further complicating detection.
2. Additionally, since SAM is used without fine-tuning, it tends to produce smooth masks that may lack precise, fine-grained

Table 7
Comparison of the performance of different segmentation models on different defect types.

Models	Crack	Spalling	Corrosion
UNet	Struggles with tiny/ thin cracks	Capture the main regions, but sometimes struggle with irregular spalling boundaries	Suitable for uniform rust areas; less robust for complex texture or background, or multi-scale metallic defects Struggles with very small or disconnected defects
DeepLabV3+	Performs well but may slightly smooth over thin cracks when noise overlaps. Struggles under a complex background	Effective even in overlapped/multi-instance spalling, but struggles with very small spalling defects	Effective, but over-segmentation is visible (some background regions are misclassified) for a few images
Segformer	Identifies even thin cracks, but struggles with irregular cracks under large background noise	Detects sharp boundaries and is robust to complex background	Handles complex background more robustly and captures both small and large region defects
YOLO-seg	May miss thin/low contrast regions, struggles with irregular cracks, under large background noise	Strong pixel-wise accuracy, but slight under-segmentation of very small spalling defects	Captures main defect regions but misses more minor details.
Mask R-CNN	Robust in all scenarios, but edges can be jagged.	Over segments in case of closer region defects and struggles for very small spalling defects	Detects disconnected defect parts well, but misses some smaller rust areas Over-segmentation is visible (some background regions are misclassified) for a few images.
Proposed YOLO11-SAM	Small cracks are well captured due to SAM refinement. Difficulty in thin cracks with a different texture and a complex background Struggles with irregular cracks under significant background noise	Captures both small and large spalling defects, but may miss defects under low contrast conditions Difficulty in handling extreme depth or boundary defects	Handles complex background better and detects both large and small defects with good boundary accuracy. However, it overlaps two or more regions with the regions of no corrosion, in case of multiple instances of corrosion.
K-means	Misses fine cracks; needs morphological operations; misclassifies artificial artefacts as cracks.	Over splits into irrelevant patches	Very poor; struggles to recognise hue shifts and irregular surface textures.

boundaries. When YOLO11 predicts a single bounding box covering multiple nearby corrosion instances, SAM may struggle with instance separation, merging them into a single mask. Backbone down-sampling can suppress fine details, and zero-shot models inherently prioritise generalisation over pixel-perfect accuracy. Consequently, closely positioned, tiny, or texture-rich corrosion regions may be overlooked or merged with neighbouring non-defective areas,

Table 8
Comparison of different segmentation models.

Different scenarios	Model Performance
Noise	UNet: Skip connections retain fine details and help preserve boundary and texture integrity against noisy backgrounds, making it robust to noise. Mask R-CNN: Instance-level segmentation filters out background clutter due to strong feature extraction and spatial detail retention. Segformer: Transformer backbone captures long-range dependencies, offering contextual robustness and suppressing random noise. DeepLabV3+: Performs well but may slightly smooth over thin cracks when noise overlaps. YOLO-Seg: More sensitive to noise, especially in small-scale cracks. Proposed YOL011-SAM: performs reliably in the presence of noise due to strong feature extraction and spatial detail retention, but can misclassify noisy textures as defects under weak prompts.
Complex Background/Texture Variation	SegFormer: Transformer attention lets it separate defects from similar-looking textures. DeepLabV3+: Performs by leveraging multi-scale context, but is less robust than SegFormer. Mask R-CNN: May confuse textured backgrounds as multiple defect instances. YOLO-Seg: Background confusion is common (low precision). UNet: Works fine on simpler backgrounds but struggles in highly textured regions. Proposed YOLo11-SAM: Zero-shot mode may fail in highly textured concrete surfaces. C2PSA modules capture fine details and handle occlusions.
Irregularly shaped defects/ different orientations/ edge defects	UNet: Good for irregular crack shapes. Mask R-CNN: Handles irregularly shaped defects, but edges can be jagged. SegFormer: Strong at capturing defect edges due to attention-based learning. DeepLabV3+: Works well, but can miss very thin or irregular edges. YOLO-Seg: Bounding-box-driven segmentation struggles with irregular outlines. Proposed YOL011-SAM: Very strong at irregular boundaries due to oriented bounding box
Varied Lighting	SegFormer: Invariant to lighting due to strong feature abstraction. Mask R-CNN: Region-based features help maintain robustness. YOLO-Seg: Strong under lighting variation due to fast backbone training on diverse datasets. DeepLabV3+: May over-smooth shadows/ reflections. UNet: Sensitive to illumination changes. Proposed YOL011-SAM: Robust under lighting variation since segmentation is shape-driven; however, low-contrast cracks may be missed.
Boundary Delination	SegFormer: Sharp edges, especially for crack tips. UNet: Skip connections preserve local detail and edges. Mask R-CNN: Instance masks give precise object-level boundaries. DeepLabV3+: Good global segmentation, but sometimes blurs edges. YOLO-Seg: Bounding-box-driven; so boundaries are coarse. Proposed YOLO11-SAM: Can give sharp edges, but requires strong prompts; prone to under-segmentation
Computational time	Average training time per epoch (for image size of 512 × 512, considering the same batch size)

(*continued on next page*)

Table 8 (*continued*)

Different scenarios	Model Performance
	UNet: 680 s; DeepLabV3+: 500 s; Mask R-CNN:1024s; Segformer: 462 s; YoloSeg:95 s; YOLO11-SAM:120 s Inference time per image, including evaluation metrics computation UNet: 265 ms; DeepLabV3+:312 ms; Mask R-CNN:460 ms; Segformer: 210 ms; YoloSeg:95 ms; YOLO11-SAM:120 ms

reflecting a practical trade-off between automation, scalability, and fine-grained segmentation precision.

3. It can be observed from Fig. 23 that the proposed semi-supervised YOLO11, SAM, and Segformer could perform multi-class instance segmentation robustly despite a limited dataset in training and class imbalance. The models MaskR-CNN and DeepLabV3+ tend to over-segment due to the closer rust-like texture. These models are not able to recognise hue shifts and irregular surface textures to segment corrosion defects reliably.
4. It can be observed from Table 6 that the proposed semi-supervised model, i.e., hybrid YOLO11 and SAM, YOLOSeg and Segformer, achieves good performance in terms of accuracy and dice score compared to other direct supervised models. The proposed YOLO11 and SAM model achieves the highest precision and AUPRC on the corrosion dataset, demonstrating good performance compared to other tested models despite slightly lower recall.

It should be mentioned here that even though the results demonstrate reasonable performance on the considered corrosion dataset, further validation on larger and more diverse datasets is necessary to establish robustness and generalisation fully.

9. Conclusion

An ensemble framework combining a YOLO11/Oriented YOLO11-based supervised object detection model and a SAM2-based zero-shot segmentation model was developed for pixel-level mapping of various defect types. The semi-supervised model was evaluated on three defect categories, including four crack datasets, spalling (with and without exposed reinforcement), and a metallic corrosion dataset, and was compared with several widely used supervised and unsupervised segmentation models. Cross-dataset evaluation on crack defects was conducted to assess the generalisation capability of the proposed segmentation framework. The results demonstrated that the semi-supervised segmentation model is effective for all considered defect types. Cross-dataset evaluation further indicated that the model possesses notable generalisation capability and that cross-dataset training can significantly enhance segmentation performance. The semi-supervised model outperformed the direct supervised models and offered advantages in data annotation by utilising region-level instead of pixel-level labelling. The study concluded that the proposed Oriented YOLO11 and SAM models with box-based prompts constitute an efficient approach for segmenting cracks with varying orientations, textures, and shading conditions, as well as spalling defects (with or without exposed reinforcement) and metallic corrosion.

The proposed YOLO11–SAM model achieved precision and recall values of 0.893 and 0.759 for cracks, 0.848 and 0.833 for spalling, and 0.702 and 0.534 for corrosion, with corresponding Dice scores of 0.820 (95 % CI: 0.793–0.867) for cracks, 0.803 (95 % CI: 0.652–0.870) for spalling, and 0.606 (95 % CI: 0.519–0.713) for corrosion. These results highlight notable differences in segmentation performance across defect types. For cracks, the high precision and Dice score combined with the lower recall indicate a conservative detection behaviour, where most predicted cracks were correctly identified, while some true instances were missed. Spalling detection exhibited balanced performance, with both precision and recall remaining high, reflecting effective identification with minimal misclassification. In contrast, corrosion segmentation remained challenging, with lower precision, recall, and Dice score, likely due to substantial variability in appearance, localized or overlapping defect patterns, complex textures, and background noise. Overall, the findings illustrate the trade-offs between accuracy and coverage, providing insight into the model's strengths and limitations across different defect categories and indicating areas requiring further improvement for more robust and generalisable defect detection.

Tables 7 and 8 present the comparative performance of the proposed model alongside several existing models across different datasets and evaluation scenarios. The YOLO11–SAM and SegFormer models were the best-performing approaches for the three defect types considered in this study. It should be noted that the reported performance reflects the characteristics of the datasets used and may vary when evaluated on larger or more diverse datasets.

The proposed hybrid YOLO11 and SAM framework demonstrated strong potential for transfer learning to other defect detection tasks beyond civil infrastructure, broadening the impact of the work. By leveraging zero-shot segmentation and region-level prompts, the model was able to adapt to previously unseen defect types without requiring extensive pixel-level annotations, making it highly versatile for new domains. For example, surface defects in manufacturing, such as scratches, dents, or material inconsistencies, often share similarities with cracks in terms of localized patterns, texture variations, and occlusions; the model's capability to detect irregular and partially occluded defects can be directly applied. Similarly, geological crack detection in rock formations or soil surfaces, where defects vary widely in size, orientation, and visual appearance, can benefit from the model's multi-scale feature extraction, attention-guided segmentation, and robustness to background variability. This transferability is further enhanced by the modular design of YOLO11 for object localisation and SAM for zero-shot segmentation, allowing minimal retraining or prompt adjustments for new defect classes. Consequently, the approach provides a scalable and cost-effective solution for diverse defect detection challenges across multiple industries, reducing annotation burden, accelerating deployment, and enabling rapid adaptation to evolving inspection requirements. Further research will concentrate on improving the suggested framework's efficiency and adaptability to other defects, such as delamination and multi-class defect segmentation. The proposed framework can be extended towards quantifying damage by pixel-based metrics for different types of defects.

CRediT authorship contribution statement

Gauthami Vijayakumar Kuttuva: Writing – original draft, Investigation, Formal analysis, Data curation. **J. Prawin:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Data availability

Data will be made available on request.

References

[1] Y.C. Chen, R.T. Wu, A. Puranam, Multi-task deep learning for crack segmentation and quantification in RC structures, Autom. Constr. 166 (2024) 105599, https://doi.org/10.1016/j.autcon.2024.105599.

[2] N. Manakitsa, G.S. Maraslidis, L. Moysis, G.F. Fragulis, A review of machine learning and deep learning for object detection, semantic segmentation, and human action recognition in machi7ne and robotic vision, Technologies 12 (2024) 15, https://doi.org/10.3390/technologies12020015.

[3] J. Long, E. Shelhamer, T. Darrell, Fully convolutional networks for semantic segmentation, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2015) 3431–3440, https://doi.org/10.1109/CVPR.2015.7298965.

[4] O. Ronneberger, P. Fischer, T. Brox, U-net: convolutional networks for biomedical image segmentation, Med. Comp. Comp. Assisted Intervent. 18 (2015) 234–241, https://doi.org/10.1007/978-3-319-24574-4_28.

[5] J. Prawin, A.U. Murad, Structural defect classification with pre-trained inception network: insights from sample knowledge and model transfer learning on four benchmark datasets, J. Struct. Eng (Madras) 52 (2025) 24–37, https://doi.org/10.5281/zenodo.17405831.

[6] J. Deng, W. Dong, R. Socher, L.J. Li, K. Li, L. Fei-Fei, ImageNet: a large-scale hierarchical image database, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2009) 248–255, https://doi.org/10.1109/CVPR.2009.5206848.

[7] C. Han, H. Yang, T. Ma, S. Wang, C. Zhao, Y. Yang, CrackDiffusion: a two-stage semantic segmentation framework for pavement crack combining unsupervised and supervised processes, Autom. Constr. 160 (2024) 105332, https://doi.org/10.1016/j.autcon.2024.105332.

[8] C. Han, H. Yang, Y. Yang, Enhancing pixel-level crack segmentation with visual mamba and convolutional networks, Autom. Constr. 168 (2024) 105770, https://doi.org/10.1016/j.autcon.2024.105770.

[9] Z. Zou, S. Yang, M. Wang, B. Song, Multilevel retentive networks with edge enhanced attention for infrastructure crack detection, Eng. Struct. 343 (2025) 121191, https://doi.org/10.1016/j.engstruct.2025.121191.

[10] Z. Zhou, J. Zhang, C. Gong, Hybrid semantic segmentation for tunnel lining cracks based on Swin transformer and convolutional neural network, Comput. Aided Civ. Inf. Eng. 38 (2023) 2491–2510, https://doi.org/10.1111/mice.13003.

[11] B.A. Kyem, J.K. Asamoah, E. Denteh, A. Danyo, A. Aboah, Self-supervised multi-scale transformer with attention-guided fusion for efficient crack detection, Autom. Constr. 181 (2026) 106591, https://doi.org/10.1016/j.autcon.2025.106591.

[12] F. Yuan, Z. Zhang, Z. Fang, An effective CNN and transformer complementary network for medical image segmentation, Pattern Recogn. 136 (2023) 109228, https://doi.org/10.1016/j.patcog.2022.109228.

[13] Y. Liu, Y. Chen, L. Jiang, X. Wang, Z. Lin, CrackFormer: Transformer network for fine-grained crack detection, in: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2021, pp. 15790–15799, https://doi.org/10.1109/ICCV48922.2021.01554.

[14] H. Fukuoka, T. Fujiu, Deep learning and transformers in bridge inspection: a review, Autom. Constr. 141 (2023) 104486, https://doi.org/10.1016/j.autcon.2022.104486.

[15] T. Wan, W. Zhao, L. Zhang, Transformer-based approach for bridge crack detection, Eng. Struct. 275 (2023) 115140, https://doi.org/10.1016/j.engstruct.2022.115140.

[16] Y. Tang, Q. Feng, R. Li, J. Xu, Detection of wind turbine blade cracks using vision transformers, Renew. Energy 213 (2023) 118–129, https://doi.org/10.1016/j.renene.2023.06.006.

[17] H. Xiao, L. Li, Q. Liu, X. Zhu, Q. Zhang, Transformers in medical image segmentation: a review, Biomed. Signal Process. Control 84 (2023) 104791, https://doi.org/10.1016/j.bspc.2023.104791.

[18] C. Zhang, C.C. Chang, M. Jamshidi, Simultaneous pixel-level concrete defect detection and grouping using a fully convolutional model, Struct. Health Monit. 20 (2021) 2199–2215, https://doi.org/10.1177/1475921720959821.

[19] J. Qian, et al., Deep learning-based pavement crack detection with noisy and complex backgrounds, Eng. Struct. 286 (2023) 116001, https://doi.org/10.1016/j.engstruct.2023.116001.

[20] H. Xu, X. Su, Y. Wang, H. Cai, K. Cui, X. Chen, Automatic bridge crack detection using a convolutional neural network, Appl. Sci. 9 (2019) 2867, https://doi.org/10.3390/app9142867.

[21] Z. Zhang, et al., Algorithm for pixel-level concrete pavement crack segmentation based on an improved U-net model, Sci. Rep. 15 (2025) 6553, https://doi.org/10.1038/s41598-025-91352-x.

[22] V. Hoskere, Y. Narazaki, T. Hoang, B. Spencer, MaDnet: multi-task learning for structural damage detection using semantic segmentation, J. Civ. Struct. Heal. Monit. 10 (2020) 757–769, https://doi.org/10.1007/s13349-020-00406-9.

[23] C. Bai, C. Zhang, Detection of concrete cracks and spalling using mask R-CNN, Autom. Constr. 129 (2021) 103855, https://doi.org/10.1016/j.autcon.2021.103855.

[24] G. Xu, Q. Yue, X. Liu, H. Chen, Investigation on the effect of data quality and quantity of concrete cracks on the performance of deep learning-based image segmentation, Expert Syst. Appl. 237 (2024) 121686, https://doi.org/10.1016/j.eswa.2023.121686.

[25] I.J. Goodfellow, et al., Generative adversarial nets, Adv. Neural Inf. Proces. Syst. 27 (2014), https://doi.org/10.48550/arXiv.1406.2661.

[26] T. Jin, X.W. Ye, Z.X. Li, Establishment and evaluation of conditional GAN-based image dataset for semantic segmentation of structural cracks, Eng. Struct. 285 (2023) 116058, https://doi.org/10.1016/j.engstruct.2023.116058.

[27] C. Shorten, T.M. Khoshgoftaar, A survey on image data augmentation for deep learning, J. Big Data 6 (2019) 60, https://doi.org/10.1186/s40537-019-0197-0.

[28] X. Ma, Z. Fu, D. Wu, Pavement crack image synthesis using generative adversarial networks, IEEE Trans. Intell. Transp. Syst. 21 (2023) 903–923, https://doi.org/10.3934/mbe.2024038.

[29] J. Shin, S. Lee, Structural defect detection using GANs under data scarcity, Sustainability 13 (2021) 12682, https://doi.org/10.3390/su132212682.

[30] R. Dunphy, V. Hoskere, B. Spencer, Multi-class damage imagery synthesis using GANs for structural inspection, Sensors 22 (2022) 6193, https://doi.org/10.3390/s22166193.

[31] J. Kim, et al., Generative AI-driven data augmentation for crack detection in physical structures, Electronics 13 (2024) 3905, https://doi.org/10.3390/electronics13193905.

[32] S. Bang, S. Lee, Cracks and spalling detection in concrete structures using GAN-based data augmentation, Sustainability 13 (2021) 12682, https://doi.org/10.3390/su132212682.

[33] A. Kirillov, et al., Segment anything, in: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2023, pp. 4015–4026, https://doi.org/10.1109/ICCV51070.2023.00369.

[34] Y. Zhang, Z. Shen, R. Jiao, Segment anything model for medical image segmentation: current applications and future directions, Comput. Biol. Med. (2024) 108238, https://doi.org/10.1016/j.compbiomed.2024.108238.

[35] K. Yang, Y. Bao, J. Li, T. Fan, C. Tang, Deep learning-based YOLO for crack segmentation and measurement in metro tunnels, Autom. Constr. 168 (2024) 105818, https://doi.org/10.1016/j.autcon.2024.105818.

[36] L. Alzubaidi, et al., A survey on deep learning tools dealing with data scarcity: definitions, challenges, solutions, tips, and applications, J. Big Data 10 (2023) 46, https://doi.org/10.1186/s40537-023-00755-2.

[37] J.M. Goo, X. Milidonis, A. Artusi, J. Boehm, C. Ciliberto, Hybrid-Segmentor: hybrid approach for automated fine-grained crack segmentation in civil infrastructure, Autom. Constr. 170 (2025) 105960, https://doi.org/10.1016/j.autcon.2024.105960.

[38] C. Zhang, M. Bahrami, D.K. Mishra, M.M. Yuen, Y. Yu, J. Zhang, SelectSeg: uncertainty-based selective training and prediction for accurate crack segmentation under limited data and noisy annotations, Reliab. Eng. Syst. Saf. 259 (2025) 110909, https://doi.org/10.1016/j.ress.2025.110909.

[39] T. Diwan, G. Anirudh, J.V. Tembhurne, Object detection using YOLO: challenges, architectural successors, datasets and applications, Multimed. Tools Appl. 82 (2023) 9243–9275, https://doi.org/10.1007/s11042-022-13424-y.

[40] Y. An, L. Kong, C. Hou, J. Ou, An adaptable rotated bounding box method for automatic detection of arbitrary-oriented cracks, Struct. Health Monit. 23 (2024) 3312–3335, https://doi.org/10.1177/14759217231212223.

[41] Z. Chen, et al., The devil is in the crack orientation: A new perspective for crack detection, in: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2023, pp. 6653–6663, https://doi.org/10.1109/ICCV51070.2023.00613.

[42] R. Khanam, M. Hussain, YOLOv11: An overview of the key architectural enhancements, arXiv:2410.17725 (2024).

[43] I. Gohar, A. Halimi, W.K. Yew, J. See, Optimizing wind turbine surface defect detection: A rotated bounding box approach, in: Proceedings of the European Signal Processing Conference (EUSIPCO), 2024, pp. 631–635, https://doi.org/10.23919/eusipco63174.2024.10715208.

[44] D. Cheng, Z. Qin, Z. Jiang, S. Zhang, Q. Lao, K. Li, SAM on medical images: A comprehensive study on three prompt modes, arXiv:2305.00035 (2023).

[45] A. Wang, M. Islam, M. Xu, Y. Zhang, H. Ren, SAM meets robotic surgery: An empirical study on generalization, robustness and adaptation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2023, pp. 234–244, https://doi.org/10.1007/978-3-031-43904-0_23.

[46] E. Akin, H. Caltenco, K.S. Adewole, R. Malekian, J.A. Persson, Segment Anything Model (SAM) Meets Object Detected Box Prompts, in: IEEE International Conference on Industrial Technology (ICIT), 2024, pp. 1–7. Accessed Nov. 25, 2025, https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10541006.

[47] M.A. Mazurowski, et al., Segment anything model for medical image analysis: an experimental study, Med. Image Anal. 89 (2023) 102918, https://doi.org/10.1016/j.media.2023.102918.

[48] F. Liang, Q. Li, H. Yu, W. Wang, CrackCLIP: Adapting vision-language models for weakly supervised crack segmentation, Entropy 27 (2) (2025) 127, https://doi.org/10.3390/e27020127.

[49] I. Loshchilov, F. Hutter, "Decoupled weight decay regularization," arXiv: 1711.05101 (2017).

[50] P. Goyal, et al., Accurate, Large Minibatch SGD: Training ImageNet in 1 Hour, arXiv:1706.02677, 2017.

[51] L.N. Smith, N. Topin, "Super-Convergence: Very Fast Training of Neural Networks Using Large Learning Rates," arXiv:1708.07120 (2019).

[52] Ultralytics, YOLO Documentation: Model Training and Hyperparameter Tuning, Online resource. Accessed Nov. 25, https://docs.ultralytics.com/modes/train/, 2025, https://docs.ultralytics.com/guides/hyperparameter-tuning/.

[53] Baeldung, Relation Between Learning Rate and Batch Size, Online resource. Accessed Nov. 25, https://www.baeldung.com/cs/learning-rate-batch-size, 2025.